SANOS-Evolve: A Discrete Stochastic-Local-Volatility Model for European-Option Smile Dynamics*

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Version 3 draft

Abstract

European option prices identify risk-neutral marginals but not the martingale transition law connecting them. SANOS-Evolve fixes an arbitrage-free marginal surface—supplied by the SANOS framework—and calibrates a discrete stochastic-local-volatility construction to two complementary observable readouts of leading-order at-the-money smile dynamics: the skew-stickiness ratio (SSR), a normalised measure of spot–volatility coupling and its maturity decay, and a forward-variance dispersion readout carrying the amplitude that the SSR normalisation divides out. The model combines two carried volatility timescales with a finite Gaussian-mixture return kernel. A log-sum-exp normalisation makes it a martingale by construction—exactly so before the recompression that holds the component budget flat, and measured rather than guaranteed after it—while a deterministic leverage overlay aligns the conditional variance level with the fixed marginals to leading order in the step. This yields analytic one-step finite-mixture propagation and deterministic readouts.

The two targets live under different measures, and the paper states the identification restriction linking them rather than asserting an equality. The forward-variance observation is VIX at-the-money implied volatility on SPX and a corrected realised strip off index, entering as a softly weighted reading rather than a coequal joint-calibration market. Across nine SPX regimes from 2012 to 2024 one uniform protocol places the model skew-stickiness inside every year's sampling band (0.9–3.1% RMS) and fits the VIX curve to 1.8–9.2% RMS; an NDX study preserves the SSR fit but exposes a two-timescale decay limit. What is identified is a class of observationally equivalent kernels, not a unique parameter vector.

Keywords: transition-kernel identification; smile dynamics; skew-stickiness ratio; arbitrage-free marginals; martingale couplings; Gaussian-mixture kernels; skew-aware deltas.
JEL: G13; C61.

Contents

*Working paper. Comments welcome.

1 Introduction

Understanding how the implied-volatility smile evolves is a central problem in derivatives pricing and risk management. A European-option surface can be fitted accurately across strikes and maturities and still leave forward-starting prices, path-dependent exposures, and smile-aware hedges undetermined. The reason is that vanilla prices pin down the distribution of the underlying at each maturity without pinning down how it travels between them.

That gap has a precise statement. At each maturity T, under the usual regularity and no-arbitrage conditions, the call surface determines a risk-neutral marginal [3]

$$\mu_T = \mathrm{Law}_{\mathbb{Q}}(S_T).$$

Across maturities, however, the family $\{\mu_T\}$ constrains but does not determine the conditional transition law

$$K_{T_1,T_2}(x, \mathrm{d}y) = \mathrm{Law}_{\mathbb{Q}}(S_{T_2} \in \mathrm{d}y \mid X_{T_1} = x)\,. \tag{1}$$

Infinitely many martingale kernels carry the same marginal chain—the observation that underwrites the model-independent bounds of martingale optimal transport [20]. Every quantity listed above depends on which of them the market is taken to follow, and the vanilla surface does not choose. Selecting one is a second inverse problem, layered on top of static calibration.

Yet "smile dynamics" is not itself an observable calibration target. Before a parametric family is imposed, the missing transition law is infinite-dimensional, and the statement that spot and volatility move together stays qualitative until one names functionals that can both be estimated from data and evaluated on a candidate kernel. Those functionals fix two things at once: what agreement with observed dynamics is taken to mean, and how two otherwise admissible kernels are to be compared.

We therefore work with two complementary leading-order at-the-money readouts. The first is the skew-stickiness ratio, the normalised regression response of at-the-money implied volatility to a spot return,

$$\beta_T = \frac{\mathrm{Cov}\big(\Delta\Sigma_{\mathrm{ATM}}(T),\, r\big)}{\mathrm{Var}(r)}, \qquad \mathrm{SSR}(T) = \frac{\beta_T}{\overline{\mathrm{skew}}(T)}. \tag{2}$$

Its level places the smile response on the sticky map—sticky-moneyness, sticky-strike, and the local-volatility regime above them—and its term structure measures how far that coupling persists with maturity. Being a ratio, though, it is comparatively insensitive to the amplitude of volatility's own motion: to leading order in this normalisation the denominator carries the same scale as the numerator, so much of the amplitude cancels. A forward-variance dispersion readout supplies that missing scale, observed on SPX through the VIX at-the-money implied-volatility term structure, which is sensitive primarily to the dispersion and persistence of future variance. The two are coupling and amplitude coordinates rather than substitutes—each is dominated by one of them, neither is a pure measurement of it. Together they are a parsimonious and empirically checkable description of leading-order at-the-money dynamics—not a complete statistic for the transition law. Calibrating to them identifies the corresponding readout-equivalence class within the chosen kernel family, and nothing finer.

This paper takes that selection as its subject. We propose SANOS-Evolve, a discrete stochastic-local-volatility framework that treats the missing transition kernel as the object of calibration. An arbitrage-free marginal surface is supplied by the SANOS framework [1] and held fixed, while a tractable two-timescale Gaussian-mixture kernel is calibrated to those two readouts. The two layers share one representation: read as a density fit, SANOS delivers each μ_T as a finite Gaussian mixture in log-moneyness—via Black–Scholes evaluation of its basis components, restated in Appendix A—and propagating one through a Gaussian-mixture kernel K_{T_1,T_2} as in (1) returns a finite mixture again. Before the state-dependent leverage overlay the intrinsic propagation stays *exactly* inside the class; the overlay retains it through the collocation and recompression of Section 3. Either way the dynamic readouts are deterministic fixed-resolution evaluations rather than simulations. A deterministic leverage overlay aligns the conditional variance level with the prescribed marginals, leaving the kernel to describe the relative volatility states, their persistence, and their dependence on the return innovation. The resulting finite-mixture construction admits explicit propagation and deterministic dynamic readouts. In the empirical implementation the realised skew-stickiness term structure is the primary target and the forward-variance block is softly weighted beside it.

The paper develops this identification architecture (Section 2), constructs the martingale kernel (Section 3) and its readouts (Section 4), and evaluates it across SPX and NDX regimes and in a smile-roll replay (Section 5); the approximation and stability results supporting the broader mixture framework are stated where they are used and proved in Appendix B, while target construction and the calibration protocol are collected in Appendices D–E.

1.1 Kernel-first identification

Conditional on the fixed marginal surface, the transition kernel is the unknown and observed dynamic behaviour selects it (Figure 1).

The construction separates the marginal level from the relative stochastic-volatility structure. Conditional on the carried volatility state the intrinsic return law is translation-invariant in log spot—computationally valuable, but unable in general to connect an arbitrary prescribed marginal pair—so a deterministic state-dependent leverage overlay supplies the local scaling the fixed surface implies, while the intrinsic kernel keeps the dispersion, persistence and return–state dependence the readouts identify. At finite resolution the resulting marginal compatibility is measured by the digital residual rather than assumed.

The implemented kernel is a parsimonious realisation of that architecture: two carried persistence timescales and a renewed within-step return branch, with martingality imposed structurally by a log-sum-exp normalisation. Conditional on the volatility state and the branch the log-return is Gaussian, so the transition law is a finite Gaussian mixture whose weights and moments are explicit functions of the parameters, and one-step propagation of the intrinsic kernel is an analytic mixture update. Finite mixtures have a long history as a tractable smile-consistent family [12]; specifying the transition law directly rather than a driving process has an architectural analogue in Markov-functional interest-rate models [45], which share a low-dimensional Markov driver, a deterministic functional calibration map and an efficient implementation. That analogy is structural—what selects the kernel here is observed dynamic behaviour.

The readout vector implements those two coordinates and carries a third block that is reported rather than optimised. $\mathcal{S}$ is the realised skew-stickiness term structure, read at five tenors from one week to three months. $\mathcal{V}$ is the forward-variance dispersion block, whose market observation is the VIX at-the-money implied-volatility term structure on SPX and a corrected realised strip off index—related observation operators occupying one slot, not the same quantity. Only these two enter the objective. The third, $\mathcal{D}$, is a fixed digital band comparing the propagated and

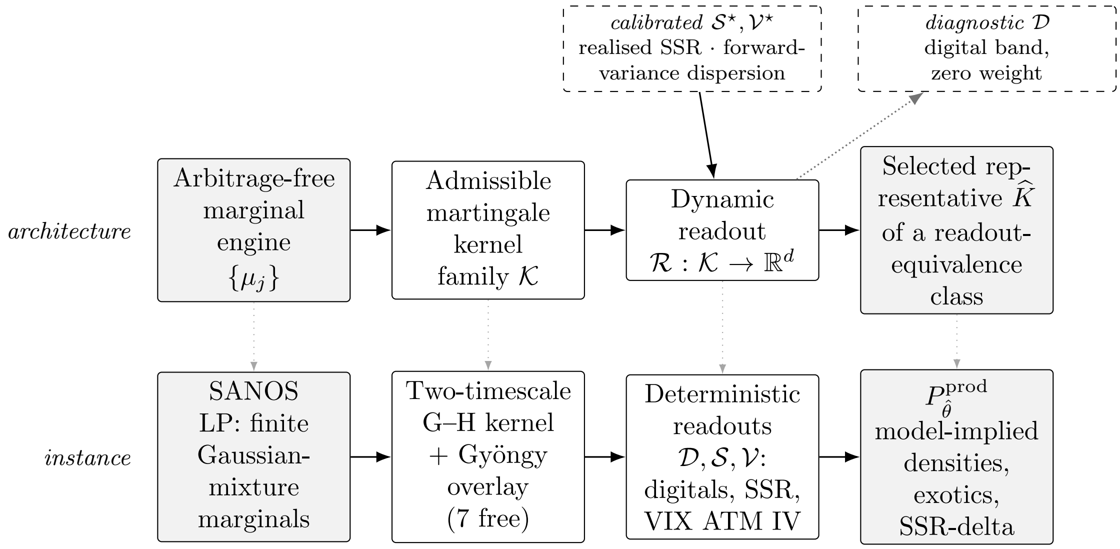

Figure 1: The identification pipeline (upper row) and the instance implemented here (lower row). The general construction needs only a convex-ordered marginal engine, an admissible martingale-kernel family $\mathcal{K}$, and a dynamic readout $\mathcal{R}$; the kernel is the unknown. Two blocks are *calibrated*—the coupling readout $\mathcal{S}$ and the amplitude readout $\mathcal{V}$—and what they select is a representative of a readout-equivalence class, not a unique kernel. The digital band $\mathcal{D}$ runs the other way: it is emitted and reported at zero loss weight (dotted), a diagnostic of the leverage overlay rather than a target. Everything to the right of the kernel is model-implied. The lower row instantiates each stage with the SANOS marginal engine, the two-timescale Gauss–Hermite kernel with a Gyöngy leverage overlay, and the deterministic digital, skew-stickiness and variance-index readouts.

target marginals: a diagnostic of the leverage overlay's finite-resolution performance, entering the loss at zero weight. Section 4 gives each observation operator, the measure each lives under, and the identification assumption linking the pooled physical-measure regression to its within-regime risk-neutral analogue.

The static layer used in the empirical implementation is SANOS [1]. It fits weights only, over a fixed grid of Gaussian bumps in log-moneyness sharing one bandwidth, each priced by Black–Scholes; its linear program enforces convex order. The fitted marginal is therefore itself a finite Gaussian mixture—a smoothed grid of point masses, which the bumps become as the bandwidth vanishes.

That representation is why this engine and not another. The marginal layer arrives in the same class the kernel propagates, so a mixture updated by a Gaussian-mixture kernel is again a mixture, and the local variance and its Breeden–Litzenberger density are closed forms on that same object. We retain those marginals and replace the canonical discrete-local-volatility transition associated with them. The architecture itself asks only for a convex-ordered marginal engine with an explicit tractable representation—arbitrage-free surface interpolation [41] and its mixture-preserving variants [46] supply alternatives—but it is the closure under propagation that keeps every readout a finite sum.

1.2 How existing methods select the kernel

Other constructions select the same missing law by other means. *Process-first* models—Heston [9], SABR [10], the Bergomi family [22, 23], rough volatility [25, 26]—specify state dynamics and derive the kernel, so smile and smile dynamics are encoded together. *Local projection* draws the dynamics from the marginals: Dupire local volatility [2] and the contemporaneous implied trees [4, 5] are the limiting case in which the static surface determines the dynamic law—so its skew-stickiness is pinned by the static skew, near the Doeff–Kamal value under an empirical $\tau^{-1/2}$ scaling [39], while the measured SPX term structure is materially maturity-dependent. *Stochastic-local volatility* [7, 8, 13] fixes a backbone and identifies a leverage function through Gyöngy's projection [6], by particle or learning-based schemes [14, 16], with existence known in the regime-switching case [17]; the leverage recovers the marginal projection, while the remaining conditional dynamics stay backbone-dependent. *Transport and entropy* methods—Bass constructions [42, 44], martingale optimal transport [43, 15], martingale Schrödinger bridges [35]—impose marginal or market-price constraints and select an admissible law by a variational or entropic criterion.

A fourth family models the observable surface *directly*: market models of implied volatility posit surface dynamics and derive the no-arbitrage drift restrictions those dynamics must satisfy [32, 33], and recent work learns the joint dynamics of liquid option prices under no-arbitrage constraints [34]. Selecting dynamics from observed surface behaviour is therefore not new; what differs is the object carried. Those models evolve the surface, so static no-arbitrage is a constraint their dynamics must respect at every date, whereas the marginals here are fixed once by a convex-ordered engine and the unknown is the kernel between them.

In each family the smile response follows from the chosen process, projection, or optimality principle, and the empirical consequence is measurable. Calibrating two-factor Bergomi, rough Bergomi, rough Heston and Heston as closely as possible to the same SPX smile, [29] finds SSR term structures that all deviate materially from the market's, and concludes that roughness alone does not change the joint spot–volatility dynamics enough to close the gap; [27] supplies the analytic counterpart, a model-free expression for the SSR whose short-time limit is $H + \frac{3}{2}$ rather than the diffusive 2. The statistic can also be targeted directly: the two-factor Quintic OU model [30] calibrates to SPX at-the-money volatility, skew and SSR term structures, and separately to the joint SPX–VIX smiles under a penalty holding the model SSR inside $[0.9, 2.0]$. One qualification shapes what is claimed below: such comparisons are drawn against point estimates of a windowed realised SSR carrying no sampling band, whereas every fit here is reported against its own displayed precision scale, which is the target's joint-HAC standard error at every date but one (Appendix C).

Adjacent joint SPX–VIX calibration. Joint SPX–VIX calibration addresses a related but different inverse problem. Guyon imposes SPX-smile, VIX-future and VIX-smile constraints and selects a minimum-relative-entropy law [37, 35]; Guo, Loeper, Obłój and Wang formulate finite SPX and VIX price constraints through semimartingale optimal transport [36]; and Che et al. derive first-order risk perturbations around an entropically calibrated SPX–VIX coupling, using the SSR to transmit SPX shocks to the VIX side [38]. SANOS-Evolve does not seek to reproduce a variance-index market. It fixes each underlying's marginal chain and calibrates a restricted explicit kernel to skew-stickiness and a softly weighted forward-variance-dispersion readout. VIX at-the-money implied volatility supplies that observation for SPX; off index, a corrected realised proxy built from the underlying's own option strip fills the same slot, and variance-index futures are not separate calibration constraints. The distinction that matters is therefore joint cross-market calibration versus portable readout-based kernel calibration, not competing solutions to one fitting problem.

1.3 Contributions and findings

The paper makes three contributions.

1. **Dynamic-readout identification of a stochastic lift.** We develop a selection principle for smile dynamics: conditional on an independently fitted marginal surface, the latent transition kernel is calibrated directly to the two complementary leading-order readouts motivated above, and what it selects is a readout-equivalence class. A deterministic leverage overlay supplies the local variance level, while the kernel controls the relative stochastic-volatility states, their persistence, and their dependence on the return innovation.

2. **An explicit finite realization and approximation foundation.** We construct a two-timescale Gaussian-mixture martingale kernel with structural fibrewise normalization, analytic one-step mixture propagation, and deterministic finite-sum SSR and variance-index readouts. Two propositions accompany the construction—leverage and finite-kernel stability (Proposition 2) and dynamic-readout regularity (Proposition 3)—proved in Appendix B.

3. **Cross-regime evidence and identified limits.** Across nine SPX regimes from 2012 to 2024, one uniform protocol places the realised SSR term structure within every year's HAC sampling band (0.9–3.1% RMS) and fits the VIX ATM implied-volatility term structure to 1.8–9.2% RMS. An NDX study using own-strip and realised forward-variance observations preserves the SSR fit but exposes a steep-decay limitation in the variance-of-variance term structure. A smile-roll replay—a residual-variance proxy, not traded profit and loss—suggests that the benefit of a smile-aware adjustment is stability relative to a trailing realised regression rather than superior short-horizon forecasting; a fixed ratio performs similarly.

The baseline is an ATM-dynamics model: it targets the level response summarized by SSR and one forward-variance distributional readout, with VIX entering as a softly weighted observation rather than as a joint calibration target. Its diffusive finite-factor structure does not reproduce rough $T \downarrow 0$ scaling, and translation invariance excludes spot-level dependence of conditional smile shape beyond the latent-regime channel. Section 6 evaluates those limits against the evidence.

2 Identification architecture and guarantees

The marginal engine supplies an arbitrage-free spot law at every listed maturity. The dynamic unknown is a transition kernel on a lifted spot–volatility state, whose role is not to refit the vanilla surface but to select a conditional coupling whose observable dynamics agree with the chosen readouts. This section defines those objects and separates the ideal admissible problem from the finite-resolution construction used in calibration.

2.1 The identification problem

Fix a pricing date and maturities $T_0 < T_1 < \cdots < T_M$ with forwards F_j and discount factors D_j, and work in log-moneyness $z = \log(S/F)$. Four objects.

A fixed marginal chain. An external convex procedure returns arbitrage-free spot laws $\{\mu_j^Z\}_{j=0}^M$, increasing in convex order, with an explicit finite mixture representation. They are fixed once and are not re-fitted when the dynamic targets change.

A lifted source law. The kernel acts on a lifted state $X_j = (Z_j, U_j)$ carrying a latent volatility regime, with law $\widehat{\mu}_j(\mathrm{d}z, \mathrm{d}u)$ whose spot projection is the engine's output, $\mathrm{pr}^Z_{\#}\widehat{\mu}_j = \mu_j^Z$. What is transported from one maturity to the next is $\widehat{\mu}_j$; the engine pins only its projection. A spot marginal cannot be pushed through a kernel whose source carries a latent volatility state.

The ideal family, and the implemented object. The *ideal* family $\mathcal{K}$ is the set of Markov kernels $\mathcal{K}_j(x, \mathrm{d}y)$ on the lifted state that are positive, normalised, martingale in the forward-normalised spot on every fibre, and—together with a deterministic leverage overlay—propagate μ_j^Z to μ_{j+1}^Z. The implementation does not sit in it, so the implemented object is not a kernel at all. It is a *law-evolution pipeline* P_θ^{prod}: an overlaid finite kernel applied to a mixture, followed by the recompression projection of Section 3.4, which maps laws to laws and is not a Markov kernel. The pipeline is indexed by $\theta \in \Theta_{\mathrm{prod}}$, the box of Section 5. Section 2.2 says what P_θ^{prod} retains of admissibility and what it does not.

A dynamic readout. A map $\mathcal{R}$ returning the finite vector of quantities the calibration actually measures, defined on a kernel or on a pipeline alike. Identification is the inverse problem

$$\widehat{K} \in \arg\min_{K\in\mathcal{K}} \big\| W\big(\mathcal{R}(K) - \mathcal{R}^\star\big)\big\|_2^2, \tag{3}$$

in the ideal statement, over kernels, and

$$\widehat{\theta} \in \arg\min_{\theta\in\Theta_{\mathrm{prod}}} \big\| W\big(\mathcal{R}_{\mathrm{cal}}(P_\theta^{\mathrm{prod}}) - \mathcal{R}^\star_{\mathrm{cal}}\big)\big\|_2^2 \tag{4}$$

as implemented—over *parameters* indexing the pipeline, and over the calibrated blocks only, since $\mathcal{D}$ carries zero weight. Every fitted object in this paper solves (4); (3) is the problem it approximates. Here $\mathcal{R}^\star$ is the observed targets and W a fixed weighting; Section 4 writes $\mathcal{R}$ out in full. Figure 1 sets this architecture beside the instance implemented here and against the standing answers to the selection question.

2.2 Ideal and finite-resolution admissibility

Definition 1 (Admissible kernel)**.** *Fix the lifted source law $\widehat{\mu}_j(\mathrm{d}z, \mathrm{d}u)$ of Section 2 and the target spot marginal μ_{j+1}^Z. A Markov kernel $\mathcal{K}_j$ on the lifted state is* admissible *if*

$$\mathcal{A}_j\big(\widehat{\mu}_j, \mu_{j+1}^Z\big) = \Big\{\mathcal{K}_j :\ \mathcal{K}_j \ge 0,\ \mathcal{K}_j\mathbf{1} = 1,\ \mathbb{E}[e^{Z_{j+1}} \mid X_j = x] = e^{Z_j}\ \forall x,\ \mathrm{pr}^Z_\#\big(\widehat{\mu}_j\mathcal{K}_j\big) = \mu_{j+1}^Z\Big\}. \tag{5}$$

The conditions are, in order: positivity, normalisation, the (forward-normalised) martingale property, and exact propagation of the spot *marginal.*

Definition 2 (Finite-feature marginal compatibility)**.** *The production chain is not an element of $\mathcal{A}_j$, and we do not claim membership of a Wasserstein ball around it. Two of the four conditions in* (5) *hold outright—positivity and normalisation—and the martingale condition holds in the weakened form*

$$\mathbb{E}\big[e^{Z_{j+1}} \mid \mathcal{G}_j\big] = e^{Z_j}, \tag{6}$$

with $\mathcal{G}_j$ the σ-algebra the implementation conditions on (the component index and the price abscissa of (23)*) and* before *the recompression projection of Section 3.4, which preserves mass and the unconditional forward only. The fourth condition, exact propagation of the spot marginal, is replaced not by a metric bound but by a* finite-feature discrepancy*: the seven-threshold digital vector of Section 4.2, together with the call-price and implied-volatility residuals at the quoted strikes. These are what Section 5 reports.*

The natural metric would be W_1 on the forward-normalised marginals, which in one dimension is

$$W_1(\mu,\nu) = \int_0^\infty \big|F_\mu(K) - F_\nu(K)\big|\,\mathrm{d}K = \int_0^\infty \big|\partial_K C_\mu(K) - \partial_K C_\nu(K)\big|\,\mathrm{d}K, \tag{7}$$

the second equality because $\partial_K C_\mu(K) = F_\mu(K) - 1$ for undiscounted calls—so W_1 is the total variation across strikes of the call-price difference. A finite set of quoted strikes does not resolve that total variation, so we report the discrepancy we can measure and make no claim about W_1.

Theorem 1 (No static arbitrage)**.** *Let $\{\mu_j\}_{j=0}^M$ be probability measures on $(0,\infty)$ with $\mathbb{E}_{\mu_j}[S] = F_j$. Assume their forward-normalised laws $\tilde{\mu}_j = \mathrm{Law}(S_{T_j}/F_j)$ are increasing in convex order, $\tilde{\mu}_0 \preceq_{\mathrm{cx}} \tilde{\mu}_1 \preceq_{\mathrm{cx}} \cdots \preceq_{\mathrm{cx}} \tilde{\mu}_M$. If $\mathcal{K}_0, \ldots, \mathcal{K}_{M-1}$ are admissible (each $\mathcal{K}_j \in \mathcal{A}_j$), then for any initial* lifted *law $\widehat{\mu}_0$ with $\mathrm{pr}^Z_\# \widehat{\mu}_0 = \mu^Z_0$, the lifted chain $\widehat{\mu}_{j+1} = \widehat{\mu}_j \mathcal{K}_j$ satisfies $\mathrm{pr}^Z_\# \widehat{\mu}_j = \mu^Z_j$ for every j, and e^{Z_j} is a martingale in the lifted filtration—so the discounted underlying is a martingale with one-dimensional marginals exactly $\{\mu_j\}$. The induced call surface $C_j(K) = D_j \int (s-K)^+ \mu_j(\mathrm{d}s)$ is then free of strike (butterfly) and calendar static arbitrage. Moreover $\mathcal{A}_j \neq \emptyset$ for every j.*

Proof sketch. Strike no-arbitrage. For each j, μ_j is a probability measure with $\mathbb{E}_{\mu_j}[S] = F_j$; hence $C_j(K) = D_j \int (s-K)^+ \mu_j(\mathrm{d}s)$ is convex and non-increasing in K with the correct bounds, which is equivalent to absence of butterfly arbitrage [3].

Martingale and marginals. Admissibility gives $\mathbb{E}[e^{Z_{j+1}} \mid X_j = x] = e^{Z_j}$ for every lifted state x, so e^{Z_j} is a martingale in the lifted filtration. It also gives $\mathrm{pr}^Z_\#(\widehat{\mu}_j \mathcal{K}_j) = \mu^Z_{j+1}$; propagation happens on the lifted law, $\widehat{\mu}_{j+1} = \widehat{\mu}_j \mathcal{K}_j$, and only its projection is pinned. Starting from $\mathrm{pr}^Z_\# \widehat{\mu}_0 = \mu^Z_0$, induction gives $\mathrm{pr}^Z_\# \widehat{\mu}_j = \mu^Z_j$ for all j, so the spot marginals are exactly $\{\mu_j\}$ and the discounted underlying is a martingale.

Calendar no-arbitrage. A martingale transition implies, by conditional Jensen, that the forward-normalised marginals are increasing in convex order, $\tilde{\mu}_j \preceq_{\mathrm{cx}} \tilde{\mu}_{j+1}$, which is equivalent to absence of calendar arbitrage for the call surface.

Existence. By Strassen's theorem [19, 20] a martingale kernel q on the forward-normalised *spot* coordinate with $\mu^Z_j q = \mu^Z_{j+1}$ exists iff $\tilde{\mu}_j \preceq_{\mathrm{cx}} \tilde{\mu}_{j+1}$, and SANOS enforces that convex order across maturities. Strassen returns a spot object, so it must be lifted before it is an element of $\mathcal{A}_j$. Disintegrate the coupling as $q_z(\mathrm{d}z')$, let the transition ignore the current latent label, and attach any probability law $\eta_{z'}(\mathrm{d}u')$ for the next one:

$$\mathcal{K}_j\big((z,u), \mathrm{d}(z',u')\big) = q_z(\mathrm{d}z')\, \eta_{z'}(\mathrm{d}u'). \tag{8}$$

This is non-negative and normalised; it is fibrewise a martingale, since $\int e^{z'} q_z(\mathrm{d}z') = e^z$ for every u; and for any $\widehat{\mu}_j$ projecting to μ^Z_j, $\mathrm{pr}^Z_\#(\widehat{\mu}_j \mathcal{K}_j) = \mu^Z_j q = \mu^Z_{j+1}$. Hence $\mathcal{A}_j \neq \emptyset$. □

The content is that the no-arbitrage *claim attaches to the admissible set*, and that the set is nonempty precisely because SANOS delivers marginals in convex order (Strassen). The discrete-local-volatility transition associated with the SANOS parameterisation [1] becomes an element of $\mathcal{A}_j$ once embedded in the lifted state—for instance with latent labels drawn independently of the return—and it is the rigid element the introduction set out to replace. The two calibrated readouts of Section 4 *select a different, controllable element* of $\mathcal{A}_j$ in the ideal statement, and they never replace the set. What is fitted is not that element: the production pipeline is not in $\mathcal{A}_j$ at all, since it satisfies the martingale condition at its own numerical state and before recompression, and replaces exact marginal propagation by the finite-feature discrepancy of Definition 2. $\mathcal{A}_j$ is the ideal object the construction targets, not a set the implementation is claimed to sit in.

One question remains before any particular kernel is written down: is the class we are about to restrict to rich enough to be worth restricting? The useful form of that question is not density among admissible martingale kernels but whether the family's *readout image* covers the behaviour the market shows—a finite-dimensional question, answered by the attainable-range experiment of Section 5.1 and the nine-regime panel of Section 5.3. Table 1 records which layer every later claim attaches to.

2.3 Structural guarantees and approximation layers

Each layer of the construction carries a property that holds exactly, by structure, and a separate element whose accuracy is finite. Table 1 grades them, and every later use of "exact" names the row it belongs to.

Layer	Structural property	Finite-resolution element
Marginal engine	arbitrage-free finite-mixture surface, convex-ordered	quote and interpolation error
Kernel	positivity and normalisation, at every state and parameter value	none
Martingality	exact at the numerical state (23), *pre-projection*	recompression preserves mass and the unconditional forward only
One-step update	analytic finite Gaussian-mixture propagation	the chosen discrete kernel
Local-variance level	the continuous relation (21) matched to leading order	the remainder R^{c}_{Δ} of (24), measured
Carried factor law	the branch rule is a fixed quadrature, exact in its own moments	Gaussian closure: the marginal law is a finite mixture, propagated by its moments
Marginal compatibility	nothing structural: the overlay targets it, no limit theorem is claimed	finite-resolution digital discrepancy
Readouts	deterministic fixed-resolution evaluation of the implemented finite functional	central difference, Newton inversion, quadrature and interpolation

Table 1: Structural guarantees and finite-resolution layers. The middle column holds exactly by construction; the right column is measured and reported.

What the construction avoids, relative to classical stochastic-local volatility, is the continuous-time leverage fixed point.

3 A finite martingale realization

We now construct one tractable member of the admissible architecture. The design has four steps. A finite branch kernel represents the joint return–state transition; a two-timescale coefficient map supplies persistence and leverage; a structural normalization enforces martingality; and a deterministic local-variance overlay aligns the conditional variance level with the fixed marginal surface. The resulting kernel remains explicit enough to propagate as a finite Gaussian mixture.

3.1 A finite martingale kernel

The state is the lifted pair of Section 2, written concretely as

$$X_j = (Z_j, U_j), \qquad Z_j = \log(S_{T_j}/F_j), \quad U_j = \big(u_j^{\mathsf{F}}, u_j^{\mathsf{S}}\big) \in \mathbb{R}^2, \tag{9}$$

with Z_j log-spot and U_j two *continuous* latent volatility factors, each standardised to unit stationary variance. The class of kernels we work in is

$$\mathcal{K}\big((z,u),(\,\mathrm{d}z',\,\mathrm{d}u')\big) = \sum_{\ell=1}^{n_{\mathsf{L}}} w_\ell\,\mathcal{N}\big(\,\mathrm{d}z';\,z+d_\ell(u),\,V_\ell(u)\big)\,Q_\ell\big(u,\,\mathrm{d}u'\big), \tag{10}$$

a finite mixture over a within-step branch index ℓ of Gaussian return increments, each branch carrying its own Gaussian transition Q_ℓ of the two factors. Three properties of (10) do the structural work, and the rest of the paper leans on them repeatedly.

Translation invariance of the intrinsic return law. The spot enters only as the additive centre of the Gaussian, so the conditional return law—and hence the conditional smile in moneyness—depends on the factor pair alone, not on the spot level. This is what makes the intrinsic propagation an exact mixture map, before the marginal overlay reintroduces a z-dependence (Section 3.4), and the dynamic readouts deterministic finite sums (Section 4).

The branch index mediates return–factor dependence. A common ℓ appears in both the Gaussian mean and the factor transition Q_ℓ, so the joint law of return and next factor state is not a product of its marginals. This is precisely the dependence that leverage and skew-stickiness measure, and it is why an approximation theorem for this class must control the joint conditional law rather than the two conditional marginals separately (Appendix B).

Gaussian destination factor law. Each $Q_\ell(u, \cdot)$ is Gaussian, so a law carried as a Gaussian mixture in (z, u) maps to another one and every readout reduces to a finite *quadrature* sum—not to a sum over a latent alphabet, of which there is none.

The two carried factors and the within-step branch play different roles, and the distinction matters throughout:

Index	Persisted across steps?	Economic role
u^{F}	yes (continuous)	fast volatility factor; the decaying short end of the skew-stickiness term structure
u^{S}	yes (continuous)	slow volatility factor; the sustained long-end floor
ℓ	no (renewed each step)	within-step return branch; carries the instantaneous skew and the return–factor coupling

Table 2: Roles of the two carried factors and the within-step branch. Sans-serif $\mathsf{F}, \mathsf{S}, \mathsf{L}$ name them. Only the branch ℓ carries a node count, n_{L}: the two persistent factors are continuous and are propagated as Gaussian moments rather than on abscissas (Section 3.4), so they have no node count of their own.

Spot marginal versus lifted law. The kernel acts on the joint state, so its one-step law is a lifted joint $\widehat{\mu}_j(\mathrm{d}z, \mathrm{d}u)$, whereas the marginal engine supplies only the spot marginal $\mu_j^Z(\mathrm{d}z) = \int_{\mathbb{R}^2} \widehat{\mu}_j(\mathrm{d}z, \mathrm{d}u)$. The first maturity is lifted with the stationary bivariate factor law π of (17); thereafter the joint is carried by the propagation. Throughout, $\mu_j \mathcal{K}_j = \mu_{j+1}$ abbreviates the match of *spot* marginals after propagation of the lifted law, $\mathrm{pr}^Z_\#(\widehat{\mu}_j \mathcal{K}_j) = \mu^Z_{j+1}$.

Equation (10) admits unrestricted weights, drifts and variances; the implementation below restricts them to an eight-coefficient image, seven of them fitted. Two scope points belong here rather than in the appendix. Appendix B establishes stability and readout continuity for the overlaid finite kernel, and *no* approximation-theoretic claim is made for the family: what that image reaches is answered by measurement (Sections 5.1 and 5.4), not by a density theorem. And the two operations the production chain performs at *every* step—evaluating the leverage overlay by collocation at finitely many price abscissas (Section 3.3), and recompressing the propagated mixture back to a fixed component budget (Section 3.4)—appear in no result of that appendix, which is stated throughout at a fixed numerical pattern (Remark 1). Their effect is measured (Section 5) rather than bounded.

3.2 The two-timescale specialization

The volatility state is a baseline plus a *fast* and a *slow* mean-reverting factor, the two-factor Bergomi picture [22, 23]. Conditional on that state the one-step log-return is Gaussian, but its mean and variance are functions of a continuous latent state, so the kernel is an integral over the factors rather than a finite mixture. The remedy is quadrature— applied to the factors' *effect* on the log-variance, not to the factors themselves, which stay continuous throughout. An n-point Gauss–Hermite rule represents $\mathcal{N}(0,1)$ by nodes ζ_l and weights w_l reproducing its moments through order $2n-1$,

$$\mathbb{E}_{\mathcal{N}(0,1)}[g] = \int g(\zeta)\,\frac{e^{-\zeta^2/2}}{\sqrt{2\pi}}\,\mathrm{d}\zeta \;\approx\; \sum_{l=1}^{n} w_l\, g(\zeta_l) \qquad (\text{exact for } \deg g \le 2n-1), \tag{11}$$

so replacing each continuous latent quantity by its nodes turns the integral into a finite sum, each node a volatility scenario. The nodes, base log-weights $\eta^{\mathsf{F}}, \eta^{\mathsf{S}}$ and within-step weights $w_l = \mathrm{softmax}(\eta^{\mathsf{L}})$ are *fixed* quadrature constants.

Eight structural knobs $\boldsymbol{\theta} = (\nu_{\mathsf{F}}, \nu_{\mathsf{S}}, \nu_{\mathsf{L}}, \lambda_{\text{skew}}, \rho_{\mathsf{F}}, \rho_{\mathsf{S}}, \kappa_{\mathsf{F}}, \kappa_{\mathsf{S}})$ determine every coefficient in (10); the level $\bar{\gamma}$ is a ninth coefficient but is *solved* rather than fitted, from the reference-variance condition (18) below.

Each carried factor is a standardised Gaussian AR(1),

$$u^{\mathsf{F}}_{j+1} = \kappa_{\mathsf{F}} u^{\mathsf{F}}_j + \sqrt{1-\kappa_{\mathsf{F}}^2}\,\varepsilon^{\mathsf{F}}_{j+1}, \qquad u^{\mathsf{S}}_{j+1} = \kappa_{\mathsf{S}} u^{\mathsf{S}}_j + \sqrt{1-\kappa_{\mathsf{S}}^2}\,\varepsilon^{\mathsf{S}}_{j+1}, \tag{12}$$

with standard normal innovations, so each has unit stationary variance and κ is *exactly* the one-step autocorrelation, $|\kappa| < 1$. This is the *continuous-shock idealisation.* Production replaces the branch part of each innovation by the finite Gauss–Hermite rule of (11): conditional on a branch the factor step is Gaussian, but marginally the carried law is a finite Gaussian *mixture*, and every "Gaussian" statement below—the bivariate stationary law of (17), the k-step law of (40)—is that idealisation, realised by matching moments and closing the recursion in the Gaussian family. The closure is a quadrature approximation, not an identity, and its error is *not* measured here: the refinement study of Section 5.2 varies node counts and the recompression budget, which is a different comparison. Quantifying the closure would mean propagating the full mixture against the moment-closed recursion, and we do not report it. The leverage is a correlation between those innovations and the branch variable ζ^{L}_ℓ that drives the return,

$$\mathrm{corr}\big(\varepsilon^{\mathsf{F}}_{j+1}, \zeta^{\mathsf{L}}_\ell\big) = \rho_{\mathsf{F}}, \qquad \mathrm{corr}\big(\varepsilon^{\mathsf{S}}_{j+1}, \zeta^{\mathsf{L}}_\ell\big) = \rho_{\mathsf{S}}, \tag{13}$$

leaving each factor an idiosyncratic innovation variance $(1-\kappa^2)(1-\rho^2)$. This is the return–factor channel of (10): a branch carrying a large negative return also tilts both factors, and it does so through the *innovation* rather than through log-spot.

The two factors enter the return law only through the scalar combination $x_j = \nu_{\mathsf{F}} u^{\mathsf{F}}_j + \nu_{\mathsf{S}} u^{\mathsf{S}}_j$, so the branch variance and the within-step skew tilt are

$$V_\ell(u) = \exp\big(\bar{\gamma} + x_j + \nu_{\mathsf{L}}\,\zeta^{\mathsf{L}}_\ell\big)\Delta_j, \qquad (\text{two-factor variance}), \tag{14}$$

$$\widetilde{m}_\ell(u) = -\tfrac{1}{2} V_\ell(u) + \lambda_{\text{skew}}\,\zeta^{\mathsf{L}}_\ell\,\sqrt{V_\ell(u)}, \qquad (\text{instantaneous mean-tilt / skew}), \tag{15}$$

the drift d_ℓ of (10) following from $\widetilde{m}_\ell$ by the martingale lock (19). Only x_j is integrated by quadrature, at n_{X} nodes; the factors themselves are never placed on abscissas. One derived quantity recurs throughout: the *full* one-step increment variance at a factor state,

$$\overline{V}(u) = \mathbb{E}_\ell\big[V_\ell(u)\big] + \mathrm{Var}_\ell\big(\widetilde{m}_\ell(u)\big), \tag{16}$$

the law of total variance applied across branches—the within-branch term plus the mean spread the skew tilt creates. It is what the level condition, the leverage overlay and the forward-variance readout all measure.

Under (12)–(13) the stationary factor law is bivariate normal with unit marginals and correlation

$$c = \frac{\sqrt{(1-\kappa_{\mathsf{F}}^2)(1-\kappa_{\mathsf{S}}^2)}\;\rho_{\mathsf{F}}\rho_{\mathsf{S}}}{1-\kappa_{\mathsf{F}}\kappa_{\mathsf{S}}}, \tag{17}$$

writing $s_\bullet = \sqrt{1-\kappa_\bullet^2}$ for the innovation scales. The only channel coupling the two factors is their shared leverage on the return branch: if either ρ vanishes they are stationarily independent. The level is then fixed by requiring the stationary expected one-step variance to match the reference,

$$\mathbb{E}_\pi\left[\overline{V}\right] = \sigma_{\mathrm{ref}}^2\,\Delta_j, \tag{18}$$

which determines $\bar\gamma$ and removes it from the fitted vector.

The parameters separate by role: $\nu_{\mathsf{F}}, \nu_{\mathsf{S}}$ set the *spread* of the log-variance across the two factors and ν_{L} its within-step dispersion; λ_{skew} the *return skew*; $\kappa_{\mathsf{F}}, \kappa_{\mathsf{S}}$ the *persistence*; and $\rho_{\mathsf{F}}, \rho_{\mathsf{S}}$ the *return–factor dependence*, the innovation leverages. A down move tilts the factors toward higher volatility through $\rho_{\mathsf{F}}, \rho_{\mathsf{S}}$. Because the leverage enters the innovation around a *fixed* stationary target rather than shifting that target, the equilibrium volatility is not a deterministic function of spot: this is genuine stochastic volatility, and the kernel stays translation-invariant in z.

Since κ is an autocorrelation it carries an effective rate directly, through the one-step half-life $-\log 2/\log\kappa$, and needs no separate interpretation. Across the nine SPX fits of Section 5.3 the ordering $\kappa_{\mathsf{S}} > \kappa_{\mathsf{F}}$ holds at every date, with median half-lives of 0.7 weeks fast and 52 weeks slow, so the two timescales the construction posits are the two the calibration finds.

The finite approximation is not a discretisation of the two persistent factors. They are carried continuously—each step propagates their means and covariance, and the state passed between steps is (weight, μ, σ, m_{F}, m_{S}, V_{FF}, V_{SS}, V_{FS}) per component—so what is finite is the quadrature that integrates their *combined* effect on the log-variance, the within-step branch count n_{L}, the price sub-abscissas, and the recompression budget. Production values and the measured readout sensitivity are reported with the empirical protocol in Section 5.

3.3 Structural martingality and the marginal overlay

Martingality is enforced twice: once on the intrinsic kernel, and again after the market-level overlay that rescales its variance. The first is stated for the *ideal* construction, at a fixed latent state; the second is what the implementation performs, and it conditions on the numerical state instead. The difference is spelled out below and is not cosmetic.

Stage 1: the intrinsic kernel (ideal). The within-step drift is locked to the forward on *every* fibre—the kernel's conditional law at a fixed state (z,u), with $u = (u^{\mathsf{F}}, u^{\mathsf{S}}) \in \mathbb{R}^2$ the continuous factor pair. Writing $t_\ell(u) = \lambda_{\mathrm{skew}}\zeta_\ell^{\mathsf{L}}\sqrt{V_\ell(u)}$ for the skew tilt of (14), so that the raw mean is $\widetilde{m}_\ell(u) = -\frac{1}{2}V_\ell(u) + t_\ell(u)$, subtract the per-fibre constant

$$A(u) = \log\sum_\ell w_\ell \exp\!\big(\widetilde{m}_\ell(u) + \tfrac{1}{2}V_\ell(u)\big) = \log\sum_\ell w_\ell\, e^{\,t_\ell(u)}, \qquad d_\ell(u) = \widetilde{m}_\ell(u) - A(u), \tag{19}$$

so that $\mathbb{E}[e^{Z_{j+1}} \mid z,u] = e^z$ exactly at a realised u (Proposition 4). The shift is a single per-fibre constant, independent of z, stable and differentiable, and it preserves the relative branch tilts that carry the leverage and the skew-stickiness.

Stage 2: the market-level overlay. A translation-invariant kernel cannot, on its own, carry a prescribed marginal chain: its unleveraged propagation has convolution form $\mu_j \mathcal{K}_j = \mu_j * \eta_j$, which connects only marginal pairs admitting a common increment-law representation. Matching is restored by one deterministic ingredient, a leverage overlay $\sigma_{\mathrm{LV}}(z, T_j)$ that rescales the conditional variance. Writing the increment as a local scale times a unit stochastic-volatility factor,

$$\operatorname{Var}\big(\Delta z \mid z, u\big) = \underbrace{\sigma^2_{\mathrm{LV}}(z, T_j)\,\Delta_j}_{\text{level (market)}} \cdot \underbrace{\nu_u}_{\text{shape (kernel), } \mathbb{E}_\pi[\nu]=1} + R^u_\Delta, \qquad \nu_u = \frac{\bar{V}_u}{\mathbb{E}_\pi[\bar{V}]}, \tag{20}$$

with $\bar{V}_u = \overline{V}(u)$ of (16): the first two factors are the *target* decomposition, exact in the continuous-time limit, and R^u_Δ is what the weekly step leaves *at a fixed latent state.* The implementation does not condition there, so a second remainder R^c_Δ—same algebra, coarser conditioning—appears at (24) below and is the one measured; the two are not the same number, and only R^c_Δ is reported. Taking $\mathbb{E}_\ell[V_\ell]$ alone mis-scales the leverage by $\sim 3\times$ at the calibrated λ_{skew}. The overlay itself is the finite-kernel analogue of the stochastic-local-volatility leverage relation,

$$\sigma^2_{\mathrm{LV}}(z, T_j) = \frac{\sigma^2_{\mathrm{Dupire}}(z, T_j)}{\mathbb{E}[\nu \mid z]}, \tag{21}$$

with σ_{Dupire} read from the fixed marginals: a closed-form Gaussian-mixture call, a Breeden–Litzenberger density, and a calendar finite difference. That difference is well posed only on a convex-ordered surface, which is what the constraint in Algorithm 1 is for; the overlay inherits that hypothesis from the marginal engine (Assumption 1). Rescaling the variance changes $\mathbb{E}[e^{\Delta z}]$, so the drift must be re-locked *after* the overlay rather than carried over. Writing $L(z) = \sigma_{\mathrm{LV}}(z, T_j)/\sigma_{\mathrm{ref}}$, the overlay scales

$$V^{\mathrm{LV}}_\ell(u, z) = L(z)^2 V_\ell(u), \qquad \widetilde{m}^{\mathrm{LV}}_\ell(u, z) = -\tfrac{1}{2} V^{\mathrm{LV}}_\ell(u, z) + L(z)\, t_\ell(u), \tag{22}$$

the variances by L^2 and the skew tilt by L, and then re-locks. Two things follow, and only the first is exact.

The forward is restored exactly, at the state the implementation carries. The production lock is taken over the branches *and* the component's own factor quadrature together,

$$A^{\mathrm{LV}}_c(z) = \log \sum_k \sum_\ell \varpi_k\, w_\ell\, \exp\Big(\widetilde{m}^{\mathrm{LV}}_\ell(x_{ck}, z) + \tfrac{1}{2} V^{\mathrm{LV}}_\ell(x_{ck}, z)\Big), \tag{23}$$

with x_{ck} the component's factor nodes and ϖ_k their weights (Section 3.4), so that $\mathbb{E}[e^{Z_{j+1}} \mid c, z] = e^z$ identically (Proposition 5). That is martingality at the *numerical information state*—the component index and the price abscissa—which is the filtration the propagation actually carries. It is not the same as martingality conditional on a realised u: the two coincide only where the component's factor law is degenerate.

The conditional variance is matched to leading order, not exactly. Because variances scale by L^2 while the tilt scales by L, the branch dispersion of the drift does not scale as L^2. At the state the implementation conditions on—the component c and the price abscissa z of (23)—

$$\operatorname{Var}_{\mathcal{K}^L}\big(\Delta Z \mid c, z\big) = L(z)^2\, \overline{V}_c + R^c_\Delta, \qquad R^c_\Delta = \frac{L^4 - L^2}{4} \operatorname{Var}_{k\ell}\big(V_\ell\big) - \big(L^3 - L^2\big) \operatorname{Cov}_{k\ell}\big(V_\ell, t_\ell\big), \tag{24}$$

both terms vanishing at $L = 1$, with the moments taken over the *joint* factor-node/branch law $\varpi_k w_\ell$ that the lock normalises over. The same algebra at a fixed (z, u) gives the same expression with branch-only moments, but that is a finer information set than the implementation uses and understates the discrepancy by roughly a factor of two: R^u_Δ and R^c_Δ are different quantities and only the latter is reported. Since $V_\ell = O(\Delta)$ and $t_\ell = O(\Delta^{1/2})$ the covariance term dominates and both remainders are $O(\Delta^{3/2})$, so the *relative* error is $O(\Delta^{1/2})$—at a weekly step that is not automatically negligible, and Section 5.2 measures it rather than bounding it. Equation (21) is therefore the continuous-time relation the weekly construction targets, and the finite step realises it to leading order. The re-lock of (23) shifts every branch by one constant, so it restores the forward without touching R^c_Δ. The complete per-step pipeline is

$$\boxed{\text{intrinsic branch law} \longrightarrow \text{var. scaling} \longrightarrow \text{martingale re-lock} \longrightarrow \text{overlaid kernel}}$$

The division of labour. This is the decoupling that organises the paper, and it is stated here once, with the moment it applies to named. *The marginal engine determines the local variance level; the intrinsic kernel determines the relative regime dispersion and the return–regime dependence.* The overlay is a variance rescaling, so what it carries is the second moment: the per-step enforcement is (21) followed by the martingale re-lock (23), which fixes a variance and a mean and constrains nothing beyond them. The propagated *skew* is therefore not carried by the overlay but generated by the kernel, through λ_{skew} and the innovation leverages, and Section 5.3 measures how far short of the target marginal's skew it falls. The π-average of (20) equals $\sigma^2_{\text{LV}}\Delta_j$ up to the finite-step remainder of (24), so the level is the leverage's responsibility and the kernel enters only through the unit-mean ratio ν_u. Positivity and normalisation are structural in the coefficient map and martingality holds at the numerical state before recompression; marginal compatibility is the overlay's; and calibration therefore fits only dynamic parameters. The kernel's own level $\bar{\gamma}$ nearly divides out of ν and is reset by σ_{LV}, leaving it close to inert; what survives is the *relative* structure of the factor state—the $\nu_{\mathsf{F}}, \nu_{\mathsf{S}}$ spread and, through (12) and (13), the autocorrelations and the innovation leverages—which is exactly what the two calibrated readouts interrogate jointly: the leverages and the spread set the coupling that $\mathcal{S}$ measures, the spread and the autocorrelations set the amplitude and persistence that $\mathcal{V}$ measures.

The overlay is a composition of three maps—conditional variance, leverage ratio, overlaid kernel—and each must be stable for approximation of a target-compatible limit to transfer to the finite-mixture implementation.

Proposition 2 (Leverage and finite-kernel stability)**.** *Let the spot-conditional variances converge in L^2, $m_n \to m$—which Assumption 1(iv) supplies from convergence of the source laws, and which we assume rather than derive, since conditional expectations are not Lipschitz in adapted topologies [21] without further structure. Let the regularised local-variance input converge, $a_n \to a$, and the coefficient vectors converge, $\theta_n \to \theta$. Then, on the compact nondegenerate region of Appendix B, the leverage ratio $\ell_n^2 = a_n/m_n$, and the overlaid finite kernel $\mathcal{K}_{\theta_n,\ell_n}$ all converge, the last in integrated conditional Wasserstein distance. The hypothesis on θ_n is not decorative: the bound of Proposition 6 is $C_\theta\|\theta - \tilde{\theta}\| + C_\ell\|\ell - \tilde{\ell}\|_{L^2(\mu)}$, so a coefficient sequence that does not settle gives a kernel sequence that does not either, however well behaved the source laws and the local variance are.*

Proof idea. Stability of the conditional mean is not derived here—it is Assumption 1(iv), for the reason given there. Granting it, the ratio is stable because ν is bounded away from zero on the nondegenerate region; and the overlaid kernel depends on (θ, ℓ) through locally Lipschitz coefficient maps, a locally Lipschitz log-sum-exp re-lock, and a Gaussian-mixture W_1 bound. The proof is Proposition 6, at the fixed numerical pattern stated there.

Here $\mathbb{E}[\nu \mid z]$ is the *spot-conditional* average of the unit factor, distinct from the stationary $\mathbb{E}_\pi[\nu] = 1$ and tilted only by the spot–volatility correlation. For marginal matching—each step re-seeded from the target law—it is close to 1 and the residual is discretisation rather than bias. Under multi-step concatenation from a fixed spot (forward densities, hedging) it is not: the negative leverage tilt correlates high-ν regimes with low z, so $\mathbb{E}[\nu \mid z{=}0] \approx 0.7$ on SPX, and ignoring it under-reads the accumulated ATM level by $\sim$10% in volatility over a multi-month chain. The full spot-conditional correction of (21) is therefore retained for absolute-level outputs.

3.4 Propagation and component control

One-step propagation of the intrinsic kernel is exact. Translation invariance means the discretised kernel adds, conditional on a regime, a *fixed* Gaussian increment—constant in the integration variable—and fixed Gaussians convolve in closed form:

$$\mathcal{N}(\,\cdot\,; m, v) \;*\; \mathcal{N}(\,\cdot\,; d_\ell(x), V_\ell(x)) \;=\; \mathcal{N}\big(\,\cdot\,;\, m + d_\ell(x),\; v + V_\ell(x)\big). \tag{25}$$

A law carried per component as a Gaussian in (z, u) therefore propagates to a finite mixture of such Gaussians,

$$\rho_{j+1}(\,\mathrm{d}z',\,\mathrm{d}u') = \sum_c \omega_c \sum_{k,\ell} \varpi_k\, w_\ell\, \mathcal{N}\big(\,\mathrm{d}z';\, m_c + d_\ell(x_{ck}),\, v_c + V_\ell(x_{ck})\big) \otimes \mathcal{N}\big(\,\mathrm{d}u';\, \mathrm{A}\,\bar{u}_c + \mathrm{b}_\ell,\, \Sigma_\ell\big), \tag{26}$$

where ϖ_k are the n_{X} Gauss–Hermite weights integrating that component's own log-variance combination $x = \nu_{\mathsf{F}} u^{\mathsf{F}} + \nu_{\mathsf{S}} u^{\mathsf{S}}$—itself Gaussian, since the component is—and $(\mathrm{A}, \mathrm{b}_\ell, \Sigma_\ell)$ is the affine factor update of (12)–(13), with $\mathrm{A} = \mathrm{diag}(\kappa_{\mathsf{F}}, \kappa_{\mathsf{S}})$ and the branch-dependent mean shift carrying the leverage. The coefficient variability lives entirely in the outer sum over (c, k, ℓ) and never inside a convolution, so propagation of the *intrinsic* kernel is an exact finite Gaussian-mixture update, needing no state-space integration, PDE solve or Monte Carlo.

The production chain is not that object, and the difference should not be collapsed. The leverage overlay of Section 3.3 scales the branch variances by $L(z)^2$, and that z-dependence is exactly what breaks (25): fixed Gaussians convolve, z-dependent ones do not. Each overlaid step is therefore closed by a first-order collocation in z rather than by the identity above. What survives exactly is structural—positivity and normalisation at every state and parameter value, and martingality at the numerical state (23)—while the local-variance match carries the remainder (24) and marginal propagation is approximate; Table 1 grades which is which. The collocation residual is moreover a property of the overlay rather than of the component budget: the Gyöngy match is re-imposed at every step, so this is a first-order approximation error and not a quantity that refining the recompression drives to zero. (A Monte-Carlo cross-check showed the same collocation biases the multi-step skew-stickiness when it is applied to the intrinsic branch law instead.)

Why the component count must be controlled. Equation (26) multiplies the component count by $n_{\mathsf{L}} n_{\mathsf{X}} n_{\mathsf{P}}$ at each step, where n_{X} is the quadrature over the combined factor state and n_{P} the price sub-abscissas of the leverage evaluation, so composing $\mathcal{K}_j \circ \mathcal{K}_{j+1} \circ \cdots$ over J steps—needed for forward densities, exotics and intermediate maturities—would leave $(n_{\mathsf{L}} n_{\mathsf{X}} n_{\mathsf{P}})^J$ components. A single kernel application needs no reduction; concatenation does.

Recompression, and what it does *not* preserve. Each composed step therefore ends with a deterministic reduction that holds the budget flat: bands on the two factor means and then on the price mean, and within each cell a match of mass, mean and variance. It is a *law-level projection*, not a kernel operation, and the distinction matters. A cell merges descendants originating in different source states, and the re-lock that follows is a single scalar per chain, chosen so that the reduced mixture's forward equals the pre-reduction mixture's. What the projection preserves is therefore mass and the *unconditional* forward; it does not preserve the forward conditional on each preceding numerical state, and the reduced object is consequently not itself a martingale kernel. We therefore claim no Strassen or convex-order property along the recompressed chain: exact martingality is a statement about the pre-projection kernel (23) only, and what the chain carries past the projection is monitored rather than guaranteed. Restoring the pathwise statement would need conditional recompression—a per-source-cell re-lock—which the production path does not perform. Multi-step composition therefore stays tractable as an exact mixture update followed by a deterministic reduced-mixture approximation whose law and martingale residuals are measured and reported. The per-step pipeline is *propagate* $\to$ *recompress* (budget and global re-lock) $\to$ *leverage* $\to$ re-lock; the recompression *is* the projection step. The reduction is never applied across the carried factor bands.

Intermediate maturities. Because the carried factors are Ornstein–Uhlenbeck, the *latent* transition operators form a semigroup, $P_s^U P_t^U = P_{s+t}^U$ with mean-reversion rate $a = -\log \kappa/\Delta$; the overlaid and recompressed production chain does not. Writing the latent log-variance as $x_t = g_0(t) + X_t^{\mathsf{F}} + X_t^{\mathsf{S}}$ separates a time-inhomogeneous forward-variance level g_0 from a time-homogeneous generator, so one interpolates the one-dimensional curve g_0 and constructs intermediate kernels by evaluation, with no matching optimisation. The carried volatility state is what makes the construction compose at all; a spot-only kernel, marginalising the volatility state at the intermediate step, does not, which is why a memoryless kernel needs a matching optimisation.

3.5 Summary of the construction

Property	Mechanism
positivity and normalisation	mixture weights and the softmax branch rule
martingality at the numerical state	log-sum-exp lock (23) over branches and factor quadrature
marginal variance level	conditional leverage overlay (21)
fast/slow memory	the two carried AR(1) factors, $\kappa_{\mathsf{F}}, \kappa_{\mathsf{S}}$
spot–volatility comovement	shared branch ℓ in the return and in both factor innovations
tractable propagation	translation-invariant Gaussian increments (25)
bounded complexity	deterministic law-level recompression

Table 3: What each structural feature of the construction delivers.

4 Dynamic readouts and identification

Section 1 motivates $\mathcal{S}$ and $\mathcal{V}$ as complementary coupling and amplitude readouts of leading-order at-the-money dynamics. This section is the formal counterpart: it defines their observation operators, names the measure each lives under, states the calibration map, and gives the equivalence that map induces. One point carries over and is worth stating once as a design principle rather than a claim—the readout is not a summary of the kernel but the topology in which it is identified, so a family is rich enough, and an approximation good enough, only relative to the functionals one intends to read off it.

4.1 The identification map

Block	Observation	Measure	Kernel feature primarily constrained	Baseline role
$\mathcal{D}$	marginal digitals	$\mathbb{Q}$	compatibility of the leverage overlay	diagnostic
$\mathcal{S}$	realised skew-stickiness term structure	$\mathbb{P}$ target, $\mathbb{Q}$ analogue	return–factor leverage and its maturity decay	active
$\mathcal{V}$	VIX ATM implied vol (SPX), realised strip (NDX)	$\mathbb{Q}$ / $\mathbb{P}$	forward-variance persistence and dispersion	active, softly weighted

Table 4: What each readout block observes, under which measure, and which part of the kernel it principally constrains. The attribution is a statement about which coordinates each block is most sensitive to, not a partition: both calibrated blocks move with most of $\boldsymbol{\theta}$, as Table 12 shows.

The calibration reads these three blocks and optimises two, so two readout vectors have to be kept apart,

$$\mathcal{R}_{\text{cal}} = (\mathcal{S}, \mathcal{V}), \qquad \mathcal{R}_{\text{report}} = (\mathcal{D}, \mathcal{S}, \mathcal{V}),$$

the first what the objective sees, the second what is reported. Exact equivalence is a statement about the calibrated pair alone,

$$K \sim_{\text{cal}} K' \iff \mathcal{R}_{\text{cal}}(K) = \mathcal{R}_{\text{cal}}(K'), \tag{27}$$

an idealisation no fit attains. What a fit delivers is membership of a production tolerance class, stated on the pipeline and with the two blocks kept apart because their tolerances are not the same kind of quantity:

$$\Big\{\theta \in \Theta_{\text{prod}} : \ \big\|W_{\mathcal{S}}\big(\mathcal{S}(P_\theta^{\text{prod}}) - \mathcal{S}^\star\big)\big\| \le \varepsilon_{\mathcal{S}}, \ \ \big\|W_{\mathcal{V}}\big(\mathcal{V}(P_\theta^{\text{prod}}) - \mathcal{V}^\star\big)\big\| \le \varepsilon_{\mathcal{V}}\Big\}. \tag{28}$$

Only $\varepsilon_{\mathcal{S}}$ carries a sampling-error reading: it is set by the skew-stickiness target's estimated standard error. The forward-variance target is a single-day $\mathbb{Q}$ read with no sampling band, so $\varepsilon_{\mathcal{V}}$ is a *chosen* calibration tolerance—the discrepancy we are willing to leave—and nothing about it is statistical. The weightings $W_{\mathcal{S}}, W_{\mathcal{V}}$ are the corresponding diagonal blocks of the W introduced below. Every downstream *identification* claim is a claim about (28); densities and exotic prices are computed from one representative of it and remain representative-specific. Write $\theta \in \Theta \subset \mathbb{R}^{n_\theta}$ for the *free* parameters, $n_\theta = 7$ in production: the eight structural coefficients of Section 3 less ρ_{S}, which is pinned, and less the level $\bar{\gamma}$, which is not a coefficient of θ at all but solved from σ_{ref} by the level match of Section 3.3. The three blocks are read on grids that the design fixes but the model does not: a set $\mathcal{K}_{\mathcal{D}}$ of log-moneyness thresholds and a set $\mathcal{T}_{\mathcal{S}} = \{T_1, \dots, T_{d_{\mathcal{S}}}\}$ of tenors, both stepped at Δt, and a set $\{\tau_1, \dots, \tau_{d_{\mathcal{V}}}\}$ of forward-variance expiries. The readout vector is then

$$\mathcal{R}(\theta) = \big(\mathcal{D}(\theta),\, \mathcal{S}(\theta),\, \mathcal{V}(\theta)\big) \in \mathbb{R}^{d_{\mathcal{D}}} \times \mathbb{R}^{d_{\mathcal{S}}} \times \mathbb{R}^{d_{\mathcal{V}}}, \qquad d_{\mathcal{D}} = d_{\mathcal{S}}\,|\mathcal{K}_{\mathcal{D}}|, \tag{29}$$

of total dimension d as in (3), with the blocks

$$\mathcal{D}_{jk}(\theta) = \Pr_\theta\big(X_{T_j} > k\big), \qquad T_j \in \mathcal{T}_{\mathcal{S}},\ k \in \mathcal{K}_{\mathcal{D}}, \tag{30}$$

$$\mathcal{S}_j(\theta) = \text{SSR}^{\text{mod}}(T_j;\theta) \ \text{ from (37)}, \qquad j = 1, \dots, d_{\mathcal{S}}, \tag{31}$$

$$\mathcal{V}_i(\theta) = \sigma_{\text{ATM}}^{\text{VIX mod}}(\tau_i;\theta) \ \text{ from (43)}, \qquad i = 1, \dots, d_{\mathcal{V}}. \tag{32}$$

Only the last two enter the loss. The instance realises the production problem (4) with a diagonal W that vanishes on the $\mathcal{D}$ block and normalises the surviving residuals by their own targets, plus one term outside that form—a ridge on θ rather than on the readout—so the objective is best given explicitly. With targets $\mathcal{S}^\star, \mathcal{V}^\star$, stage anchor θ^0, per-tenor skew-stickiness weights $\varsigma \in \mathbb{R}^{d_\mathcal{S}}$ normalised to $\overline{\varsigma^2} = 1$, ridge weights $\varrho \in \mathbb{R}^{n_\theta}$ and a floor $\varepsilon > 0$, the residual vector is

$$r(\theta) = \Big(\underbrace{\varsigma_j \frac{\mathcal{S}_j(\theta) - \mathcal{S}_j^\star}{\mathcal{S}_j^\star}}_{j \le d_\mathcal{S}},\ \underbrace{w\,\frac{\mathcal{V}_i(\theta) - \mathcal{V}_i^\star}{\mathcal{V}_i^\star}}_{i \le d_\mathcal{V}},\ \underbrace{\varrho_k \frac{\theta_k - \theta_k^0}{\max(|\theta_k^0|,\, \varepsilon)}}_{k \le n_\theta}\Big), \qquad w = w_{\mathrm{vov}} \sqrt{\frac{d_\mathcal{S}}{d_\mathcal{V}}}, \tag{33}$$

and the objective is its least-squares cost,

$$\begin{aligned} J(\theta) &= \tfrac{1}{2} \big\| r(\theta) \big\|_2^2 \\ &= \underbrace{\tfrac{1}{2} \sum_{j=1}^{d_\mathcal{S}} \varsigma_j^2 \Big(\tfrac{\mathcal{S}_j - \mathcal{S}_j^\star}{\mathcal{S}_j^\star}\Big)^2}_{\text{skew-stickiness}} + \underbrace{\frac{w_{\mathrm{vov}}^2 d_\mathcal{S}}{2 d_\mathcal{V}} \sum_{i=1}^{d_\mathcal{V}} \Big(\tfrac{\mathcal{V}_i - \mathcal{V}_i^\star}{\mathcal{V}_i^\star}\Big)^2}_{\text{forward variance}} + \underbrace{\tfrac{1}{2} \sum_{k=1}^{n_\theta} \frac{\varrho_k^2 (\theta_k - \theta_k^0)^2}{\max(|\theta_k^0|, \varepsilon)^2}}_{\text{ridge}}, \end{aligned} \tag{34}$$

minimised over the box $\theta^{\mathrm{lo}} \le \theta \le \theta^{\mathrm{hi}}$. The grids, the weights $w_{\mathrm{vov}}, \varsigma, \varrho, \varepsilon$ and the box are design choices of the empirical study and are given with it in Section 5; the factor $\frac{1}{2}$ is the least-squares convention, carried so that the multi-start costs quoted in Appendix E are on the scale of (34).

Four features of (34) are choices rather than conventions. *Residuals are relative*: a given percentage miss counts the same at every tenor, though the levels differ by an order of magnitude across $\mathcal{T}_\mathcal{S}$. *The forward-variance block is normalised by its own length*: the factor $\sqrt{d_\mathcal{S}/d_\mathcal{V}}$ fixes that block's aggregate weight at $w_{\mathrm{vov}}^2 d_\mathcal{S}$ whatever $d_\mathcal{V}$ happens to be, so the cross-underlying comparison of Section 5.5 is like-for-like. *The digital block is absent*, not merely down-weighted (Section 4.2). *The ridge is not a target*: it penalises displacement from the stage anchor rather than misfit to data, scaled to each coordinate's own anchor value, and tames the railing the loosely identified directions would otherwise produce (Appendix E).

Two counting conditions follow, and they constrain the design rather than the model. Only the $\mathcal{S}$ and $\mathcal{V}$ blocks carry observations, so the data-bearing residuals number $d_\mathcal{S} + d_\mathcal{V}$ and nominal determinacy requires $d_\mathcal{S} + d_\mathcal{V} \ge n_\theta$; the ridge adds n_θ further residuals, but those are regularisation rather than observation, and can only make the optimiser well-posed on one. Identifiability asks more than counting. The parameters separate by role (Section 3), and so does the evidence: the skew-stickiness term structure is the primary evidence for the leverages and persistence loadings, the forward-variance term structure for the regime spreads. Since the kernel carries two timescales, the $\mathcal{S}$ curve is essentially a two-rate decay—a short-tenor level, two rates and a relative amplitude—so $\mathcal{T}_\mathcal{S}$ must both be large enough to resolve four shape degrees of freedom and *span* both timescales. A grid confined to tenors short relative to $-1/\log \kappa_{\mathsf{S}}$ steps leaves the slow factor constrained only through the ridge, however many points it contains.

4.2 Marginal compatibility

For a propagated mixture $\rho(\mathrm{d}x) = \sum_i W_i \mathcal{N}(\mu_i, \sigma_i^2)(\mathrm{d}x)$ the survival probability at threshold k is the finite sum $\Pr(X > k) = \sum_i W_i \,\Phi((\mu_i - k)/\sigma_i)$, evaluated on the grid $\mathcal{K}_\mathcal{D}$ against the interpolated target marginal at the corresponding maturity. The resulting $|\mathcal{K}_\mathcal{D}|$ survival probabilities per maturity report what the leverage overlay of Section 3.3 does and does not achieve at finite resolution; the grid used here is given in Section 5.2.

This block carries *zero weight* in the baseline fits: it belongs in the reported readout vector but not in the optimised loss, and serves as an acceptance readout. A configuration with a positive digital weight is used for the decoupling experiment of Section 5, where up-weighting $\mathcal{D}$ degrades the dynamic fit by more than an order of magnitude—which is itself the evidence that the statics and dynamics are separately controlled.

4.3 Skew-stickiness: a within-regime risk-neutral analogue

Three distinct objects are called "SSR" in this literature. We name them before computing anything.

Object	Where it lives	Role here
Empirical *pooled* realised-regression $\mathrm{SSR}^{\mathrm{emp}}$	Calendar-time panel under $\mathbb{P}$, no conditioning on the latent state	The calibration target.
Instantaneous / leading-order $\mathrm{SSR}^{\mathrm{inst}}$	Continuous-time asymptotics	Analytic interpretation, limiting values, and the hedge formula. Never a fitted quantity.
Finite-step *within-regime* $\mathrm{SSR}^{\mathrm{mod}}$, eq. (37)	The implemented discrete kernel, stationary regime law, under $\mathbb{Q}$	The fitted model readout. "Exact" scopes the evaluation of this functional, not equality with the pooled target.

Table 5: The three skew-stickiness objects. Every occurrence of "SSR" in this paper names one of these three.

The empirical target is the pooled regression $\mathrm{Cov}(\Delta\Sigma_{\mathrm{ATM}}(T), r)/[\mathrm{Var}(r)\,\mathrm{skew}(T)]$, estimated on the calendar panel with a single intercept; the latent volatility state is neither observed nor conditioned on. The empirical and model statistics measure the same economic channel but are not identical functionals: the former is a pooled physical-measure regression, the latter a stationary average of within-regime risk-neutral covariance and variance. The calibration imposes

$$\mathrm{SSR}^{\mathbb{Q},\,\mathrm{within}}_{\mathrm{mod}}(T) \;\approx\; \mathrm{SSR}^{\mathbb{P},\,\mathrm{pool}}_{\mathrm{emp}}(T)$$

as an identifying modelling restriction, not as an equality of functionals.

The two channels. The sums below run over the (f,s) nodes of the *stationary quadrature* of the bivariate factor law (17), with π_{fs} the corresponding Gaussian-copula weights. They are quadrature nodes of a continuous law, not states of a finite alphabet: the construction of Section 3 has none. For each such node (f,s) and maturity T the routine precomputes the propagated at-the-money volatility $\sigma(u^{\mathsf{F}}, u^{\mathsf{S}}; z, T)$ on the fixed factor–spot grid, its slope $\mathrm{skw}_{f,s}(T) = \partial_z \sigma(u^{\mathsf{F}}_f, u^{\mathsf{S}}_s; 0, T)$, the leveraged branch means and variances $(D_{fs\ell}, V_{fs\ell})$, the branch weights w_ℓ, the stationary node weights π_{fs}, and the Gauss–Hermite nodes and normalised weights (z_q, ω_q). With realised returns $r_{fs\ell q} = D_{fs\ell} + \sqrt{V_{fs\ell}}\, z_q$ and $\bar r_{fs} = \sum_\ell w_\ell D_{fs\ell}$, the factors take their own AR(1) step,

$$u^{\mathsf{F}}_{f\ell e} = \kappa_{\mathsf{F}} u^{\mathsf{F}}_f + s_{\mathsf{F}}\big(\rho_{\mathsf{F}}\zeta_\ell + \sqrt{1-\rho_{\mathsf{F}}^2}\,\varepsilon^{\perp}_e\big), \qquad u^{\mathsf{S}}_{s\ell e'} = \kappa_{\mathsf{S}} u^{\mathsf{S}}_s + s_{\mathsf{S}}\big(\rho_{\mathsf{S}}\zeta_\ell + \sqrt{1-\rho_{\mathsf{S}}^2}\,\varepsilon^{\perp}_{e'}\big), \tag{35}$$

with $(\varepsilon^{\perp}, \omega)$ a further Gauss–Hermite rule, applied once per factor because the innovations are independent *given* the shared branch ζ_ℓ. The one-step at-the-money volatility change is then

$$\Delta\sigma_{fs\ell e e' q}(T) = \sigma\big(u^{\mathsf{F}}_{f\ell e}, u^{\mathsf{S}}_{s\ell e'};\, r_{fs\ell q}, T\big) \;-\; \sigma\big(u^{\mathsf{F}}_f, u^{\mathsf{S}}_s;\, 0, T\big), \tag{36}$$

the first term read off the precomputed surface by trilinear interpolation in the two factor coordinates and the spot shift. It carries both channels: the *surface* channel, through evaluation at the realised return, and the *factor* channel, through (35), which moves which slice of the surface is read. No transition matrix between regimes appears: the factor update is the continuous one of (12), and only the surface lookup is discretised. The readout is the finite sum

$$\mathrm{SSR}_T^{\mathrm{mod}} = \frac{\mathcal{C}_T}{Q\,\bar{s}_T}, \qquad \mathcal{C}_T = \sum_{f,s} \pi_{fs} \sum_{\ell,e,e',q} w_\ell\, \omega_e \omega_{e'} \omega_q\, \Delta\sigma_{fs\ell ee'q}(T)\,(r_{fs\ell q} - \bar{r}_{fs}), \tag{37}$$

normalised by the model's own return variance and average skew,

$$Q = \sum_{f,s} \pi_{fs} \Big[\sum_{\ell} w_\ell \big(D_{fs\ell}^2 + V_{fs\ell} \big) - \bar{r}_{fs}^2 \Big], \qquad \bar{s}_T = \sum_{f,s} \pi_{fs}\, \mathrm{skw}_{fs}(T). \tag{38}$$

How far apart are the two functionals? Both the numerator and denominator of (37) centre *within* each regime, so by the total-covariance and total-variance identities they differ from the pooled statistic by the between-regime terms $\mathrm{Cov}(\mathbb{E}[\Delta\Sigma \mid I], \mathbb{E}[r \mid I])$ and $\mathrm{Var}(\mathbb{E}[r \mid I])$. Two independent measurements bound the gap. Model side: at the calibrated kernels $\eta_{\mathrm{var}}^{\mathbb{Q}} = 0.003$–$0.03\%$ and $\eta_{\mathrm{cov}}^{\mathbb{Q}} = 0.05$–$1.2\%$, so global re-centring would move $\mathrm{SSR}^{\mathrm{mod}}$ by at most 1.3%. Data side: partitioning the empirical panel by an observable proxy for the volatility state (the lagged one-month at-the-money volatility), the between-state share of the covariance reaches 4.7% in terciles and 8.4% in quintiles, moving the empirical SSR by a median 0.85% and at most 4.9% over 2012–2024, at both resolutions inside the sampling band of Appendix C. Neither addresses the *measure*: the model is a $\mathbb{Q}$-martingale whose only regime-dependent mean return is the $O(\sigma^2\Delta)$ Jensen correction, whereas the pooled empirical between-state terms can carry physical drift and variance-risk premium, and it is the leading-order invariance of quadratic covariation that licenses comparing the *within-state* channels. That protection does not extend to the higher-moment forward-variance channel, which governs the off-index results of Section 5.

What the ratio means for a delta. The instantaneous object fixes the convention the SSR values in this paper are quoted in. For a spot–volatility response coefficient R and skew $\mathcal{S}$,

$$\Delta(R) = \Delta_{\mathrm{BS}} + \mathrm{Vega} \cdot R\,\mathcal{S}/S \tag{39}$$

folds the level-induced move of the at-the-money volatility into the Black delta, with R in SSR units: $R{=}0$ is the no-roll reference and the variance-minimising choice is $R = \mathrm{SSR}_T$. For a *fixed-strike* option the smile-roll term shifts this by one unit, giving $\Delta_{\mathrm{BS}} + \mathrm{Vega}\,(\mathrm{SSR}_T - 1)\,\mathcal{S}/S$, whence sticky-strike is SSR=1 [11] and sticky-moneyness SSR=0—the reference values against which Section 5.6 places the calibrated kernel.

4.4 Forward-variance information

As Section 1 sets out, the skew-stickiness block constrains coupling but its normalisation divides out most of volatility's own amplitude. This block reads the forward-variance distribution directly; on SPX its market observation is the VIX at-the-money implied-volatility term structure, entering as a softly weighted reading rather than as a jointly calibrated market—the object of the joint SPX–VIX literature [37, 36, 38].

Because the carried factors are the Gaussian AR(1) pair of (12), the k-step law of U from any Gaussian state (m, V) today is available in closed form, with no transition matrix and no matrix power:

$$m_k^{\mathsf{F}} = \kappa_{\mathsf{F}}^k m_0^{\mathsf{F}}, \quad V_k^{\mathsf{FF}} = \kappa_{\mathsf{F}}^{2k} V_0^{\mathsf{FF}} + \big(1-\kappa_{\mathsf{F}}^{2k}\big), \qquad V_k^{\mathsf{FS}} = (\kappa_{\mathsf{F}}\kappa_{\mathsf{S}})^k V_0^{\mathsf{FS}} + s_{\mathsf{F}} s_{\mathsf{S}} \rho_{\mathsf{F}} \rho_{\mathsf{S}} \frac{1-(\kappa_{\mathsf{F}}\kappa_{\mathsf{S}})^k}{1-\kappa_{\mathsf{F}}\kappa_{\mathsf{S}}}, \tag{40}$$
$$m_k^{\mathsf{S}} = \kappa_{\mathsf{S}}^k m_0^{\mathsf{S}}, \quad V_k^{\mathsf{SS}} = \kappa_{\mathsf{S}}^{2k} V_0^{\mathsf{SS}} + \big(1-\kappa_{\mathsf{S}}^{2k}\big),$$

the unit stationary variances of (12) supplying the constants. The cross term is the *shared* branch shock accumulating over k steps: ρ_{F} and ρ_{S} load on the same ζ_ℓ^{L}, so the two factors do not decouple, and setting V^{FS} to zero would make their innovations independent while each stayed correlated with the return. The thirty-day forward volatility is the window average of the one-step increment variance, normalised by its stationary value so that σ_{ref} carries the level:

$$\Upsilon(m, V)^2 = \sigma_{\text{ref}}^2 \frac{\frac{1}{n_{\text{var}}}\sum_{k=1}^{n_{\text{var}}} \mathbb{E}\big[\overline{V}(U_k)\big]}{\mathbb{E}_\pi\big[\overline{V}\big]}, \qquad U_k \sim \mathcal{N}\big(m_k, V_k\big), \qquad n_{\text{var}} = \max\big(1, \text{round}(\tfrac{30}{365}/\Delta)\big), \tag{41}$$

with $\overline{V}$ the full one-step increment variance (16) and $\mathbb{E}_\pi$ that same functional on the stationary factor law—the normalisation of (18), so that Υ returns σ_{ref} on the stationary state by construction. Each term is evaluated by the combined-factor quadrature of (14), since V_ℓ depends on U only through $x = \nu_{\mathsf{F}} u^{\mathsf{F}} + \nu_{\mathsf{S}} u^{\mathsf{S}}$, itself Gaussian with mean $\nu_{\mathsf{F}} m_k^{\mathsf{F}} + \nu_{\mathsf{S}} m_k^{\mathsf{S}}$ and variance $\nu_{\mathsf{F}}^2 V_k^{\mathsf{FF}} + 2\nu_{\mathsf{F}}\nu_{\mathsf{S}} V_k^{\mathsf{FS}} + \nu_{\mathsf{S}}^2 V_k^{\mathsf{SS}}$. Today's state is a state choice rather than a parameter: the fast factor starts at its mean and the slow level u_0^{S} is bisected so that $\Upsilon\big((0, u_0^{\mathsf{S}}), 0\big)$ reproduces the observed spot index, under `no_grad` and therefore outside the optimisation. That consumes one observation—the spot variance index on SPX, the own-strip thirty-day variance-swap rate off it—which is spent on the state and is not among the residuals counted in Section 5.2. At the option expiry, $n_{\text{opt}} = \max(1, \text{round}(\tau/\Delta))$ steps out, the readout needs the *law* of Υ, and the propagation of Section 3.4 supplies it: each surviving component j carries a weight w_j, a log-price law $\mathcal{N}(\mu_j, \sigma_j^2)$, and a conditional factor law $\mathcal{N}(m_j, V_j)$ whose moments are (40) run from that component's own state. The factor is realised at the expiry, so the index value there is $\Upsilon(u, 0)$ at Gauss–Hermite nodes $u_{j,ab}$ of $\mathcal{N}(m_j, V_j)$: the residual variance is expanded, not passed back into (41), which would turn genuine dispersion of the index into a convexity correction on its mean. The leverage enters here as well—instantaneous variance is $\ell(z)^2 e^g$, so the thirty-day rate carries ℓ too—read across the component's own log-price nodes $z_{j,q}$. Production reads it as a root-mean-square over the variance window rather than at one slice,

$$\ell_{j,q} \;\propto\; \Big(\tfrac{1}{n_{\text{var}}} \sum_{m=0}^{n_{\text{var}}-1} \ell_{\,n_{\text{opt}}-1+m}(z_{j,q})^2\Big)^{1/2}, \tag{42}$$

normalised to unit weighted mean, because instantaneous variance is $\ell^2 e^g$ and the index averages it over $[\tau, \tau + 30\text{d}]$: what multiplies is the RMS across those slices, and consecutive tenors then share $n_{\text{var}} - 1$ of them instead of inheriting the ladder's per-week jumps. With ω_{jabq} the product of the four weights,

$$X_{jabq} = \ell_{j,q}\,\Upsilon\big(u_{j,ab}, 0\big), \quad F^{\text{mod}} = \sum \omega_{jabq} X_{jabq}, \quad C_{\text{ATM}}^{\text{mod}} = \sum \omega_{jabq}\big(X_{jabq} - F^{\text{mod}}\big)^+, \tag{43}$$

and the at-the-money Black inversion is closed form, $\sigma_{\text{ATM}}^{\text{VIX mod}} = (2\sqrt{2}/\sqrt{\tau})\,\text{erf}^{-1}\big(C_{\text{ATM}}^{\text{mod}}/F^{\text{mod}}\big)$. Neither choice is cosmetic. A readout that drops ℓ levers the skew-stickiness block but not this one, so the two blocks would then read different models. And when $\rho_{\mathsf{F}} = \rho_{\mathsf{S}} = 0$ the branch shock does not move components at all—every bit of factor uncertainty sits inside V_j—so folding

V_j in as convexity would report exactly no dispersion for a model with unchanged $\nu_{\mathsf{F}}, \nu_{\mathsf{S}}$. The market counterpart estimates the variance-index forward from a put–call parity regression over central strikes and reads the out-of-the-money smile at that forward, so the calibrated number is an at-the-money implied volatility rather than a generic dispersion parameter.

One qualification belongs with this block: it imposes unconditional forward-variance consistency only. The conditional identity a genuine joint calibration would require is exactly what reading rather than jointly fitting forgoes.

Portability. That weaker formulation is what makes the readout portable. The model side of the block does not change with the underlying: it is (43) in every case, an at-the-money implied volatility read off the kernel's own forward-variance law. What changes is the observation that fills the target slot. Where a liquid variance-index option market exists—among the underlyings here, SPX—that observation is the market counterpart of (43) term for term. NDX has no such market, so the target is the realised volatility of a constant-maturity variance-swap strip built from the underlying's own options (Appendix D, equation (51)). The two observation operators occupy the same slot in $\mathcal{R}$ with the same identification role and different measure content, so off index a residual in that block carries the operator mismatch as well as any kernel error. The consequences are taken up in Section 5.

4.5 Parameter roles and identifiability

Parameters	Principal effect
$\rho_{\mathsf{F}}, \rho_{\mathsf{S}}$	innovation leverage: level of spot–volatility comovement
$\kappa_{\mathsf{F}}, \kappa_{\mathsf{S}}$	factor autocorrelation; the maturity decay of that comovement
$\nu_{\mathsf{F}}, \nu_{\mathsf{S}}$	forward-variance dispersion
$\lambda_{\text{skew}}, \nu_{\mathsf{L}}$	within-step return skew and branch spread
$\bar{\gamma}$	variance level; solved from (18), not fitted

Table 6: Which parameters the readouts are principally informative about.

The kernel is translation-invariant (Section 3), so the conditional smile depends only on the regime pair, and skew-stickiness is generated by the *innovation* leverages rather than by any z-coupling: $\rho_{\mathsf{F}} = \rho_{\mathsf{S}} = 0$ gives no spot–volatility comovement and $\mathrm{SSR}^{\mathrm{mod}} = 0$ (sticky-moneyness), while raising them lifts the ratio through the sticky-strike value and $\kappa_{\mathsf{F}}, \kappa_{\mathsf{S}}$ shape its term structure—fast for the decaying short end, slow for the sustained floor. The placement of the leverage is a design decision with teeth: a Monte-Carlo cross-check showed that coupling the regimes to *spot* instead makes equilibrium volatility a deterministic function of spot, pinning $\mathrm{SSR}^{\mathrm{mod}} \to 2$ at long maturities with no re-tuning, whereas placing it in the innovation breaks that floor while keeping the readouts deterministic finite sums.

Proposition 3 (Dynamic-readout regularity)**.** *On the compact nondegenerate region of Appendix B, where liquid strikes are bounded away from the wings, vega is bounded below at the quoted at-the-money points, and forward variance and average skew are bounded away from zero, the readout* $\mathcal{R} = (\mathcal{D}, \mathcal{S}, \mathcal{V})$ *of* (29) *is continuous in the kernel, and in the implemented finite parameterisation it is locally Lipschitz and piecewise* C^1 *in* $\boldsymbol{\theta}$*.*

Proof idea. Each block is a finite composition of smooth maps on that region: a Gaussian-mixture CDF for $\mathcal{D}$; a matrix polynomial, a square root away from zero, and Black inversion with vega bounded below for $\mathcal{V}$; and for $\mathcal{S}$ a finite sum of bounded locally Lipschitz factors over two denominators kept away from zero. Kernel-to-readout stability then propagates along the finite maturity grid. The proof is Theorem 7, via the propagation recursion of Proposition 6.

Two consequences matter downstream. The objective is non-convex—implied-volatility inversion, the stationary law, the leverage, interpolation and the ratio structure of (37) all contribute—so the convexity available for semimartingale optimal transport [15, 43] does not transfer. And distinct kernels may produce the same finite readout vector; that equivalence is measured in Section 5, and is why the object carried downstream is the readout rather than the parameter vector.

5 Validation and empirical evidence

The evidence is organised around four questions. First, does the deterministic implementation reproduce the finite kernel it is intended to evaluate? Second, does the kernel possess a genuine dynamic degree of freedom once the marginal surface is held fixed? Third, can the resulting readouts be fitted across materially different SPX regimes? Fourth, what transfers off SPX, and what does the resulting smile response imply for a skew-aware delta?

The first two are settled on synthetic ground truth, where the generator is known and every quantity is computable both in closed form and by simulation (Section 5.1). The last two are settled on ORATS end-of-day chains (Sections 5.2–5.6). Target construction and filters are collected in Appendix D and the calibration protocol with its robustness checks in Appendix E; the body reports results.

5.1 Synthetic validation: implementation and attainable range

Two things must be checked before any market data is used, and only a known generator can check them: the first validates the computation, the second validates the economic degree of freedom the whole construction rests on. Both run at a fixed reference generator $\boldsymbol{\theta}_0$—the eight coefficients of Section 3.2 with $\bar{\gamma}$ solved and ρ_{S} at zero, calibrated once to canonical SPX term-structure targets—on a synthetic arbitrage-free chain. The decoupling experiment below instead runs at the *shipped* 2019 fit on that date's real surface.[1]

The implementation evaluates the kernel it claims to evaluate. The closed-form realised SSR sits within Monte-Carlo noise of a direct simulation of the same finite kernel ($|\mathrm{diff}| \approx 0.005$ at one month to one year; the fused SLV SSR ≤ 0.007), and the check has teeth: a source-averaged form of the same readout is off by 0.105 on a skewed target. The dynamic readouts of Section 4 are therefore a deterministic fixed-resolution evaluation of the implemented finite functional, which is what every real-data residual below is read against.

The dynamic degree of freedom is genuine but narrow. Hold the statics and ask how far the dynamics can still move. Along the steepest *static-preserving* direction—the null space of the static Jacobian holding five smile observables (ATM vol at 1m, 6m and 42 weeks, skew and curvature at 1m), Newton-held to $\leq$4 bp in the vols and $\leq$0.6% in skew at every accepted point—the 42-week SSR is freely movable only inside a band of width **0.05**, $[1.54, 1.59]$, with six of eight walk points

[1] $\boldsymbol{\theta}_0 = (\nu_{\mathsf{F}}\,1.490, \nu_{\mathsf{S}}\,0.076, \nu_{\mathsf{L}}\,1.311, \lambda_{\text{skew}}\,{-0.105}, \rho_{\mathsf{F}}\,0.388, \rho_{\mathsf{S}}\,0, \kappa_{\mathsf{F}}\,0.417, \kappa_{\mathsf{S}}\,0.610)$, the shipped 2019-06-03 fit. Reaching the long end needs a snapshot grid at $(4, 13, 26, 42)$ weeks rather than the five fitted tenors; the leverage ladder is the shipped 42-week one, so 42 weeks is the longest tenor the production configuration supports and the band is quoted there rather than at one year.

holding (Figure 2). The freedom is thus non-zero—the family is genuinely stochastic-volatility, off the local-volatility floor—and bounded: wider moves recruit the statics, which the full calibration supplies. The direction that moves it is dominated by the slow factor, $+0.78\,\kappa_{\mathsf{S}} - 0.47\,\nu_{\mathsf{S}}$, and the projected gradient has norm 0.17, so the reachable set is a thin sliver rather than a plateau. The moving direction is an amplitude–persistence combination—the sensitivity map of Table 4 appearing in the null space.

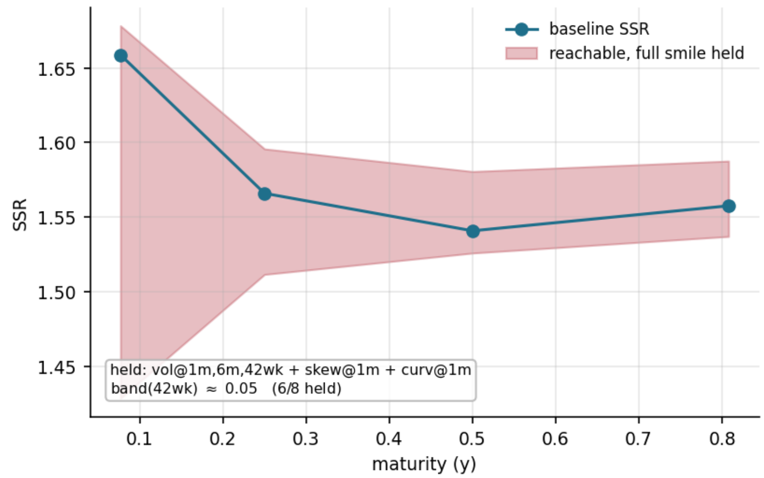

Figure 2: The statics/dynamics decoupling band at the shipped 2019-06-03 fit, computed on the construction of Section 3. Moving along the steepest static-preserving direction, the SSR is confined to the shaded band; its long end spans ≈ 1.54–1.59 at fixed statics, a width of 0.05. The band widens sharply at the short end, where no observable is held.

5.2 Data and calibration design

The real-data study calibrates one kernel per regime and reads it against two targets. Both underlyings use the same design (Table 7); the one substantive difference is the forward-variance block, risk-neutral on SPX and physical on NDX, which is the asymmetry the off-index study has to overcome.

The design constants. Section 4 leaves the readout grids and objective weights symbolic because they are choices of this study, not features of the model. Here they are. The kernel of Section 3 gives eight structural coefficients, ordered $(\nu_{\mathsf{F}}, \nu_{\mathsf{S}}, \nu_{\mathsf{L}}, \lambda_{\text{skew}}, \rho_{\mathsf{F}}, \rho_{\mathsf{S}}, \kappa_{\mathsf{F}}, \kappa_{\mathsf{S}})$, of which ρ_{S} is pinned at zero, so the free dimension is $n_\theta = 7$. The tenor grid is $\mathcal{T}_{\mathcal{S}} = \{1, 2, 4, 8, 13\}\Delta t$ with $\Delta t = 1/52$, so $d_{\mathcal{S}} = 5$ spanning one week to three months. The forward-variance block runs from six to twelve on SPX, according to how many VIX expiries survive the minimum-days filter, and is eight on NDX (14, 21, 30, 45, 60, 90, 120, 180 days). The objective uses $w_{\text{vov}} = 0.8$ scaled by $\sqrt{d_{\mathcal{S}}/d_{\mathcal{V}}}$, so the two blocks carry comparable weight whatever their lengths, with a ridge toward the stage anchor.

The counting condition is satisfied on both underlyings. On SPX the data-bearing residuals number $d_{\mathcal{S}} + d_{\mathcal{V}} = 11$ to 17 against seven free parameters; on NDX they number $5 + 8 = 13$, the realised strip being trusted at eight tenors for the reasons given in Section 5.5 and Appendix D. The off-index fit is therefore over-determined by six coordinates. That is an absence of underde-

Component	SPX	NDX
Static surface	SANOS on the selected date	SANOS on the selected date
Dynamic target	annual realised SSR panel	annual realised SSR panel
residual weighting	uniform in tenor	inverse relative HAC error
Forward-variance target	selected-date VIX ATM-IV curve	realised constant-maturity var-of-var
measure	$\mathbb{Q}$ (vol-index options)	$\mathbb{P}$ (own variance-swap strip)
observation operator	readout and target coincide	corrected; §5.5
tenors fitted	all listed VIX expiries (6–12)	eight, 14–180 d
Model maturities	five SSR tenors, 1wk–3m	same
Numerical resolution	single, fixed (Appendix E)	same
Optimisation	two-stage; accept on data cost	cold start

Table 7: Calibration design. The forward-variance block differs in measure and in observation operator, both handled explicitly in Section 5.5. The SSR residual weighting differs for a measured reason: on SPX the fitted target and the estimator behind the reported error band agree to 0.0% at all nine dates, whereas one NDX date carries a 24.9% split between them (Appendix D), so off index the residuals are weighted by their own measured reliability.

termination, not identification: it says the readouts outnumber the free parameters, not that they pin them.

Fit quality is read against the target's own precision. Each SSR target carries a joint heteroskedasticity- and autocorrelation-consistent standard error (Appendix C), HAC'd across the regression slope and the skew denominator together since they are not independent. We quote it as one number per year, the tenor-RMS of the relative standard error, and report each fit against it. A residual smaller than that standard error is descriptive: it says the miss is small relative to the target's estimated sampling variability, not that model and target have been shown equal, and not that the residual could not have been smaller. The band is a *reporting benchmark*: nothing in the optimiser consults it, and it is not a stopping rule—the fits stop on `xtol`, as Appendix E records. One row of Table 11 is an exception to the reading above: at NDX 2017 the fitted target is the Huber-robust β while the scale beside it is derived from the ordinary least-squares estimator, so for that row the number is a precision scale from a different estimator, not that target's own standard error. With seven free parameters against five SSR tenors the kernel can interpolate the point estimates outright, so the SSR block carries no positive residual degrees of freedom and admits no in-sample specification test: agreement inside the band is necessary and not sufficient. A test with positive degrees of freedom needs data withheld from the fit, which Section 5.4 does. Both underlyings are inside the band at every date in the panel.

All reported fits follow this protocol with no per-year tuning: one anchor, one schedule, one resolution, for every date and both underlyings. The protocol, the numerical resolution and its measured cost are in Appendix E; the realised-target construction and its four corrections are in Appendix D.

5.3 SPX: cross-regime identification

Finding. Across nine SPX regimes—the flattest term structure (2012), the COVID shock (2020), the bear grind (2022), the steepest low-vol surface (2024)—the kernel reproduces the realised SSR term structure *inside the year's own sampling band at every date*, and fits the VIX ATM implied-volatility curve to 1.8–9.2% RMS. Each year is a single fixed calendar date, the first June trading day.

Marginal compatibility. The statics layer and the kernel are mutually consistent before any dynamic target is imposed, and the part of that consistency which is structural should be separated from the part which is measured. *Structural.* The SANOS program is the joint linear program of Appendix A over all expiries at once, whose calendar constraint $Uq_j \geq Uq_{j-1}$ imposes convex order on the fitted marginals across $\sim$20 maturities to $\sim$2.5 y. Continuum convex order is a *standing guarantee supplied by the marginal layer*, which this paper takes as a hypothesis rather than re-derives: the program enforces it as call inequalities on a strike grid, so beyond the grid it rests on the smoothness of the mixture basis (Appendix A). The propagated kernel satisfies positivity and normalisation at every state and parameter value, and the martingale lock (23) at the numerical state the propagation carries, so the forward-start return density has mass and forward equal to 1 identically, not to within a tolerance. *Measured.* Two things hold only up to an error that is reported. The local-variance *level* is matched to leading order, its finite-step remainder (24) sitting in Table 8. And exact propagation of the next spot marginal is the one admissibility condition the production chain does not inherit, since the leverage overlay closes each step by collocation rather than by the exact Gaussian identity (26). What the chain delivers is the finite-feature marginal compatibility of Definition 2, and the condition it replaces is monitored rather than assumed: 3 call-bp and 26 IV-bp out to six months, measured after the Gyöngy match of Section 3.3 and reported in Table 8. Those are discrepancies at the quoted strikes, not a distance between laws, and no Wasserstein membership is claimed.

What the empirical sections below test is the part the construction does *not* give for free—the dynamics.

Dynamic fit. The calibration is a two-stage protocol rather than a single optimisation: a cold start, then a warm continuation from it with the calendar derivative $\partial_T C$ mollified at source, accepting the second stage only where it lowers the data cost. It is deterministic, reproducible on any new date, and never worse than the first stage by construction. Across the panel it improves both blocks—the SSR half of the objective by 32.4% and the forward-variance half by 21.6%, for 23.3% overall—and is accepted at four of nine dates (2012, 2016, 2017, 2018). It is the only intervention we tested that improves the forward-variance fit without paying for it in the SSR block.

The resulting SSR RMS averages 1.70% and is inside the target's joint-HAC standard error at all nine dates, those bands running 5.1–16.4%; the VIX ATM RMS averages 5.14% (Table 11, term structures Figures 3–4). Two qualifications belong with those numbers. The acceptance test is *in sample*—one bit per date—and at two of the nine the margin is inside the objective's own staircase noise, so the choice there is arbitrary, though those are also the dates where it costs least either way. And the forward-variance instrument is *biased*: the model's VIX law sits $\approx$11.7% low in forward while the VIX smile slopes up at $\approx$+0.6 in $\log K$, so matching ATM volatility identifies the vol-of-vol amplitude at a displaced point—a $\approx$9.1% systematic, pooled over 88 expiries, which is the same order as the entire forward-variance residual. A moneyness-matched correction was constructed and *rejected*: the displacement depends on $\boldsymbol{\theta}$, so lowering the amplitude and raising the forward are substitutes, and the correction opened a flat direction rather than closing a bias (Appendix D). The amplitude is therefore identified to within about ten percent, and we report it as such.

External calibration of the level. A published estimate places the empirical SPX skew-stickiness ratio between 0.9 and 2.0 over 2012–2022, at maturities from one month outwards and on a 60-day rolling window [30]. Over that maturity range the realised targets used here sit inside it at every tenor and every year (1.15–1.93 from one to three months); the fitted model leaves it once, at 2.03 at one month; the one-week targets run higher, 1.28–2.46, at a maturity the published band does not cover.

That comparison anchors the level rather than bounding the variation. The skew-stickiness ratio is an instantaneous ratio of covariations, so it is never observed, only estimated, and the width of any reported interval mixes market variation with the variance of the estimator behind it: the HAC standard errors of Appendix C run 4–18% of the target on the annual window used here, hence roughly twice that on a sixty-day one, so a *constant* ratio of 1.4 read through such an estimator would print about $[0.96, 1.84]$ at two standard errors—nearly the published interval itself.

The more informative comparison is not the level but whether fitting the smile delivers the dynamics. It does not. A two-factor Bergomi calibrated to the same smile does not reproduce the skew-stickiness term structure, and the two-factor Quintic OU—a model built to target smile dynamics—reaches the market range only once an SSR constraint is imposed *alongside* the smile [30]—consistent with the inverse-problem argument of Section 1: skew-stickiness enters the observed range only when it is explicitly constrained.

Stability across dates and optimizer basins. Three checks, reported in full in Appendix E. The joint objective is *loosely identified*: a cold start and a warm continuation from a neighbouring fit can differ by up to $\sim$0.6 points of SSR RMS, comparable to the fit quality being reported. That freedom is bounded and measured: across four sensible seeds the total data cost spans 0.06%, the cold start best among them, while a seed taken from a different tenor set lands one date 41% worse—which is why the protocol fixes the seed rather than searching over seeds. The parameters vary more than their fitted readouts do: any two fits the protocol accepts reproduce the same SSR and skew term structures to within the reported RMS, the residual freedom lying in the $\boldsymbol{\theta}$-directions those readouts leave soft (Table 12). Section 6 draws the consequence. This is the decoupling of Section 3.3 seen from the calibration side, and it is why the object carried downstream is the readout rather than $\boldsymbol{\theta}$. Repeating the calibration at three static dates per year over 2012–2024 moves the fitted SSR term structure by a median 7.4% at one week and 2.6–4.0% from one to three months, inside the realised-SSR sampling band the fit is judged against. And the statics/dynamics decoupling holds regime-independently *in the variance*: with the digital band given zero weight, so the kernel is fit to the dynamics alone, the from-spot leveraged marginal still matches the SANOS marginals to $\approx$5% survival RMS in calm, high-vol and grind alike—the leverage carries the variance level whatever the dynamics do. The digital band reads survival probabilities, which are integrals of the density and nearly insensitive to an at-the-money slope, and the propagated skew does fall short of the target marginal's—48% of it at one week rising to 72% at three months, on the same inversion at both sides. That is a property of the overlay rather than of the fit, since a variance rescaling constrains no third moment, and it does not enter the SSR readout, which normalises by the model's own skew.

Boundary. The two target blocks sit on different clocks—the SSR is an annual $\mathbb{P}$ regression while the marginal engine and the VIX curve are same-date $\mathbb{Q}$ cross-sections—and the static-date experiment above measures the cost of that pairing.

5.4 Held-out tenors: what the SSR block actually determines

Finding. The long end of the skew-stickiness curve is implied by the rest of the calibration; the short end is fitted. Withholding the two interior tenors costs a median 2.95% on the unseen points,

Diagnostic	Result (real SPX)	Target
Marginal discrepancy, $\mu_j\mathcal{K}_j$ vs. μ_{j+1}	3 call-bp / 26 IV-bp ($\leq$6 m)	within bid–ask
Leverage remainder δ_{LV} (24)	0.40–3.19% weighted RMS	leading-order match
Forward-start density	mass, forward = 1 to machine ε	exact
Forward-start skew	rolls down $-0.5 \to -0.14$	flatter than spot
From-spot marginal, dynamics-only fit	≈5% survival RMS	all regimes
Second-order leverage $\mathbb{E}[\nu \mid z{=}0]$	≈ 0.7; from-spot ATM +10%	—

Table 8: Architecture-level diagnostics on real SPX chains: what holds before and after the dynamic fit, independently of the per-year fit errors of Table 11. The marginal row is measured *after* the structural Gyöngy match, and is the finite-feature discrepancy of Definition 2 at the quoted strikes, not a distance between laws. The leverage row is $\delta_{\mathrm{LV}} = \mathrm{Var}_{\mathcal{K}^L}(\Delta Z \mid c, z)/\big(L^2\overline{V}_c\big) - 1$ of (24), with both moments over the joint (k, ℓ) factor-node/branch law, evaluated at the nodes the production step visits and weighted by the production weights, so it is a probability-weighted summary rather than a maximum over tail nodes; ranges are across the nine shipped fits.

below the target's own estimated standard error at eight of nine dates. Withholding the two *end* tenors costs a median 8.72%, and the damage is entirely at one week.

Design. Section 5.2 leaves the SSR block with no residual degrees of freedom; these create some. Tenors are dropped from the objective by zeroing their entries in the residual weight vector, the survivors renormalised so the retained block keeps its scale against the forward-variance block, and the model then scored where it was not told the answer. Everything else—protocol, seed, box, ridge, forward-variance targets—is the shipped configuration.

	interior: fit 1wk/1m/3m			ends: fit 2wk/1m/2m			
Year	held RMS	2wk	2m	held RMS	1wk	3m	target s.e.
2012	2.95	−1.7	−3.8	3.97	−4.8	−3.0	7.0
2016	**47.78**	**+67.6**	−0.7	8.72	−10.3	−6.9	5.1
2017	1.31	−0.1	−1.9	3.66	−2.9	+4.3	11.9
2018	5.09	−6.7	+2.7	12.10	+17.1	+0.6	10.6
2019	2.52	+3.3	−1.2	9.61	−13.6	−0.2	5.2
2020	0.94	−0.4	−1.3	13.42	−12.8	−14.0	16.4
2021	2.72	−1.2	−3.7	2.24	+3.0	+1.2	5.7
2022	3.51	+3.2	+3.8	20.45	−28.3	−5.8	6.0
2024	2.98	+3.8	−1.9	7.80	−11.0	+0.2	14.7
median	2.95			8.72			
inside s.e.	8/9			5/9			

Table 9: Held-out SSR tenors, nine SPX dates. *held RMS* is over the two withheld tenors only; the per-tenor columns are signed relative errors in percent. *target s.e.* is that year's joint-HAC standard error (Appendix C), an estimate of the target's sampling variability. Excluding 2016 the interior held-out RMS runs 0.94–5.09%.

The two ends behave differently. Splitting the ends experiment by tenor, the three-month error has median magnitude 3.0% and no sign preference (5/9 negative), while the one-week error has median magnitude 11.0% and is negative at seven of nine dates—a systematic undershoot, not noise. Predicting the long end from the interior tenors and the forward-variance curve works about as well as interpolating; predicting the short end does not. The model reaches the realised one-week

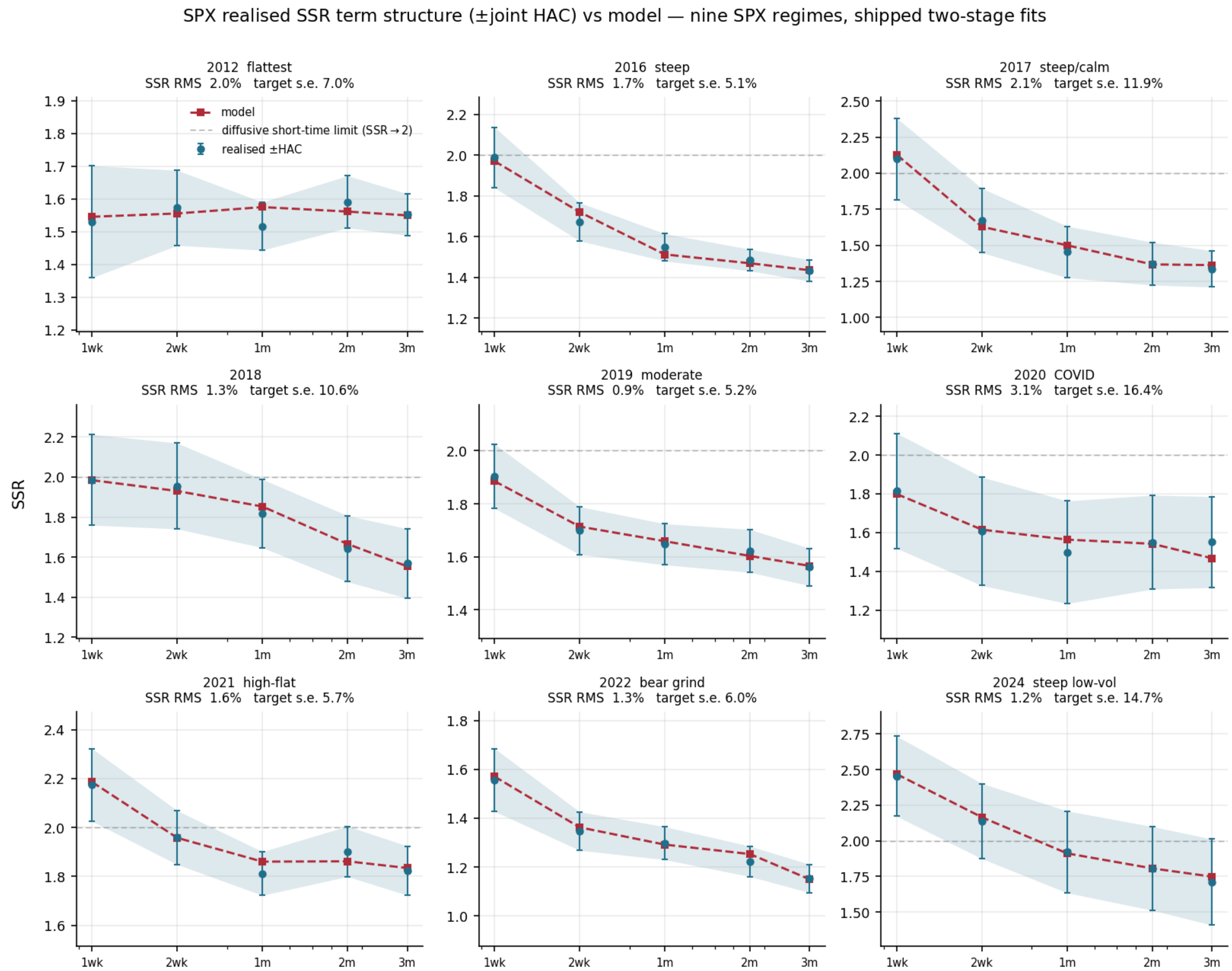

Figure 3: Realised SSR term structure vs. model, one date per year, 2012–2024. Markers, error bars and shaded band: the realised skew-stickiness (a calendar-year $\mathbb{P}$ regression) with its joint Newey–West HAC $\pm 1\,$SE, HAC'd across the regression slope and the skew denominator together since they are not independent (Appendix C). Dashed: the model SSR at the fitted $\boldsymbol{\theta}$, from the two-stage protocol of Section 5.3. Plain dashed line: the diffusive $H=\frac{1}{2}$ short-time limit SSR $\to 2$, a continuous-time benchmark rather than a bound on the finite-step statistic. **The model sits inside the band at every tenor of every date**; each panel quotes that year's RMS against its displayed precision scale, which on SPX is the target's joint-HAC standard error throughout.

value when that value is in the objective, and does not get there on its own.

Two independent results point the same way. The identifiability spectrum (Table 12) makes the slow persistence κ_{S} the stiffest direction in the problem and the fast return correlation ρ_{F} the softest; and the diffusive short end is the first limitation listed in Section 6. What the held-out test adds is a magnitude and a direction for a limitation that was previously asserted.

The 2016 *interior failure is a shape failure, not a numerical one.* With the two-week tenor withheld the model places it at 2.80 against a realised 1.67, between a well-fitted one-week (1.98) and one-month (1.60)—a hump no monotone decay would produce. It is not a numerical artefact of the ratio: (37) normalises by the model's own average skew and so diverges as that approaches zero, but the 2016 fit sits at 0.39, inside the 0.24–1.16 the panel spans and well clear of the 0.10

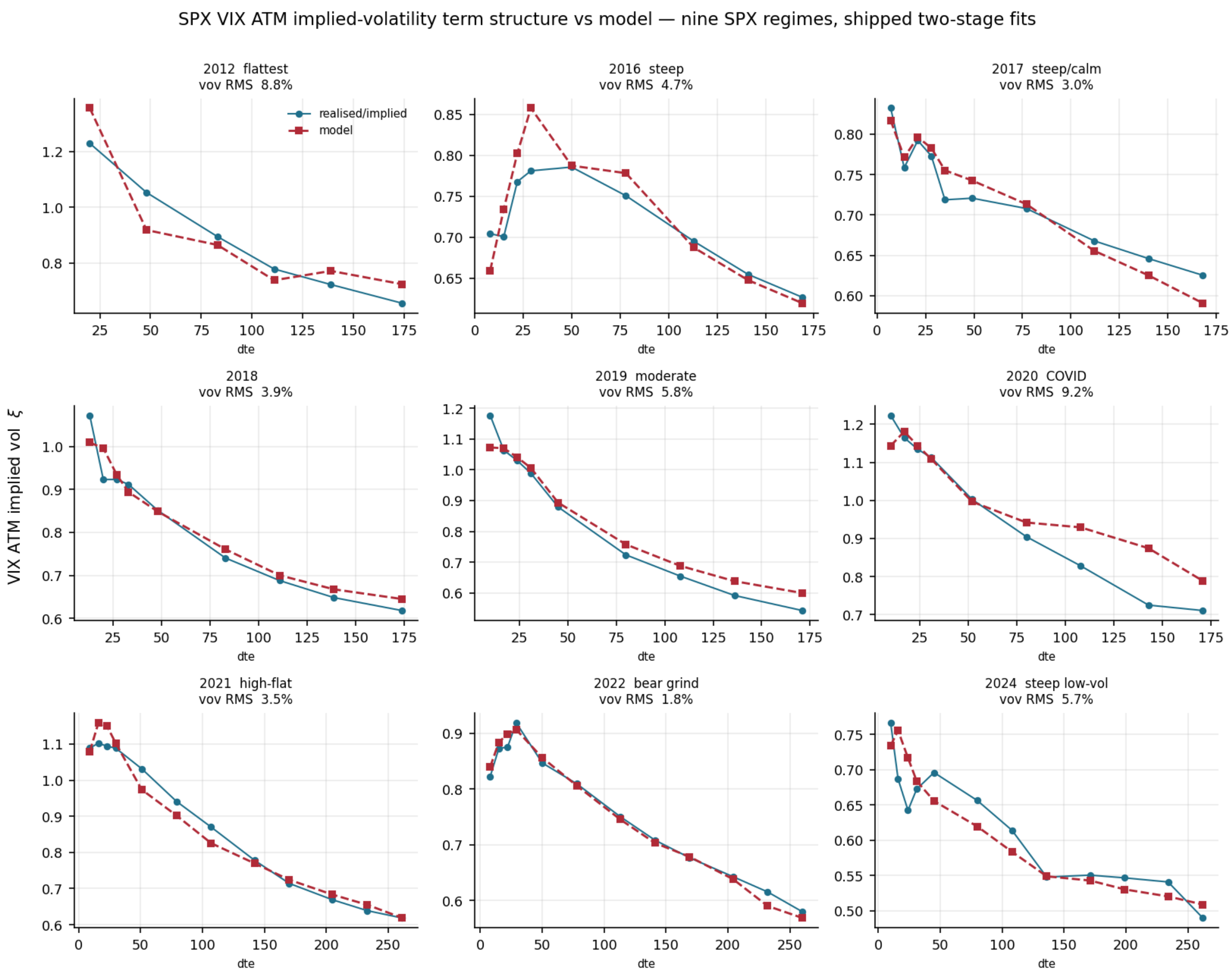

Figure 4: VIX ATM implied volatility term structure vs. model, same dates. Markers: the VIX-implied ATM volatility ξ at every listed VIX-option maturity surviving the minimum-days filter—six at 2012, twelve by 2021, so the fit is asked for more where the market provides more—a *single-day* $\mathbb{Q}$ read carrying no sampling band. Dashed: model. On this underlying the readout and the target are the same object, so no observation-operator correction is involved; the amplitude is however identified only to within the $\approx$9.1% instrument bias of Section 5.3.

at which a parameter scan finds the readout diverging. That margin is recorded per fit, which is what verifies the nondegeneracy hypothesis of Proposition 3 at the vectors the calibration visits.

Boundary. Both designs withhold tenors within a date, not dates. The forward-variance block is fully fitted throughout, the static surface is the same date's, and the withheld and retained SSR targets come from one panel of daily returns and are therefore correlated. This tests what determines the *shape* of the SSR term structure; it is not out-of-time validation, and the annual target window of Section 5.2 is untouched by it.

5.5 NDX: portability, and what the off-index target actually measures

Finding. The model travels off SPX using no volatility-index data of the underlying's own—the targets are its own option strip and realised increments—though the observation-operator correction those targets need is *estimated on SPX and transported*, which is the one imported ingredient and is quantified below. It travels on *both* blocks: the SSR RMS is 1.92%—inside the target's estimation band at all nine dates, and numerically close to SPX's 1.70% on a target built from NDX's own strip, though a residual inside one estimated standard error is a descriptive comparison and not a test—and the eight-tenor realised variance-of-variance is fitted to 14.3% against a target carrying 24% of its own roughness. The steep-decay shape gap that a nearest-expiry realised target produces is substantially an *observation-operator* artefact rather than a limit of the two-timescale decay.

The observation operator. The model's forward-variance readout (43) prices an option expiring at τ on a 30-*day* forward variance: the variance window is fixed and τ varies. On SPX the target is that same object. Off index the target must be (51), the realised volatility of the T-*day* variance-swap level, and the two are not counterparts. The gap is quantifiable on SPX, the one underlying carrying both sides, where the ratio is smooth and monotone in tenor, from 0.33 at one week through unity near three months to 1.22 at six months, with nine to ten years of coverage at every tenor. Uncorrected it is large enough to dominate: it makes the realised term structure appear to decay faster than a mean-reverting kernel can represent, and the demanded decay exponent falls from 0.44 to 0.18 once it is applied—from outside the model's reach to comfortably inside. The supporting diagnostic is that *SPX inherits the same apparent ceiling when fed realised targets*, which points at the target construction rather than at the off-index kernel without ruling the latter out.

Four corrections, two checked on held-out tenors. The realised series needs a constant-maturity construction (the nearest-expiry series rolls, and a realised volatility of a rolling series counts each roll as a move); a physical validity bound (of 25,576 observations exactly one lies outside the range an index variance-swap level can occupy, and its single increment carried 92.5% of that cell's annual variance); removal of increments spanning a bracketing-expiry change beyond three months, where listed expiries are sparse enough that the switch is a genuine discontinuity; and the observation-operator correction above. Two were checked on tenors withheld from the fit. Fitting only the 30- and 90-day anchors and scoring the seven withheld tenors, each target favours the run trained on it: the corrected-target run wins by a mean 32.5% at nine of nine dates when scored against the corrected series, and the raw-target run wins by a mean 15.2% at seven of nine when scored against the raw one. The correction is supported because the first margin exceeds the second, not because either scoring is neutral. The held-out-tenor RMS is large in absolute terms either way—31–46% across the four combinations—so this ranks two targets rather than validating a level. Fitting only 14–45 days, the churn removal predicts the held-out 60–180 day range 21.7% better—while dropping the *same number* of increments at random improves it by 0.3%, so the gain is attributable to those increments specifically and not to thinning a noisy series. Details, and two statistical screens that were built and rejected on their own pre-specified diagnostics, are in Appendix D.

What the remaining residual co-moves with. Measured as each year's deviation from its own power law—a shape the estimator has no reason to violate—the realised term structure carries 24% RMS of roughness against the model's 4.8%, a factor of five. Across the nine dates the fit residual and that roughness correlate at $R^2 = 0.93$ with an intercept near 4%: the cross-year variation in fit quality moves with the target's own smoothness. The daily coherence of the strip also degrades across the sample—the first principal component of the eight-tenor increment panel explains 96% of variance in 2012 and 46% by 2020—which is itself a reason off-index calibration is harder than the index case. Nothing here shows the off-index dynamics to be the same as SPX's; what it shows is that the instrument is measurably noisier, which is consistent with the residual gap without establishing it.

What does not dissolve. One deficiency survives every correction, and it is visible on SPX rather than NDX. Measuring each series against its own log-log shape, the model undershoots the forward-variance readout at approximately six weeks by 4.0% on SPX ($t = 3.6$ across the panel), against a traded VIX option price with no correction, transport or estimator between model and observation. The same tenor carries the largest residual on NDX (+19.6%), amplified roughly fivefold by the noisier instrument; the residual's *shape* is common to the two underlyings (correlation +0.82) while its amplitude is not. We report this as a bounded limitation of the two-timescale decay rather than repair it.

Boundary. The correction cannot be verified where it is applied, so Appendix D tests it on held-out tenors instead. The SPX control bears on it in both directions: the residual oscillation is present on SPX too, so it is not purely a transport artefact, but it is 4.4 times larger off index. Separately, the off-index marginal is modestly fat-tailed relative to the data ($\sim$2–3$\times$ the excess kurtosis), a discretisation feature.

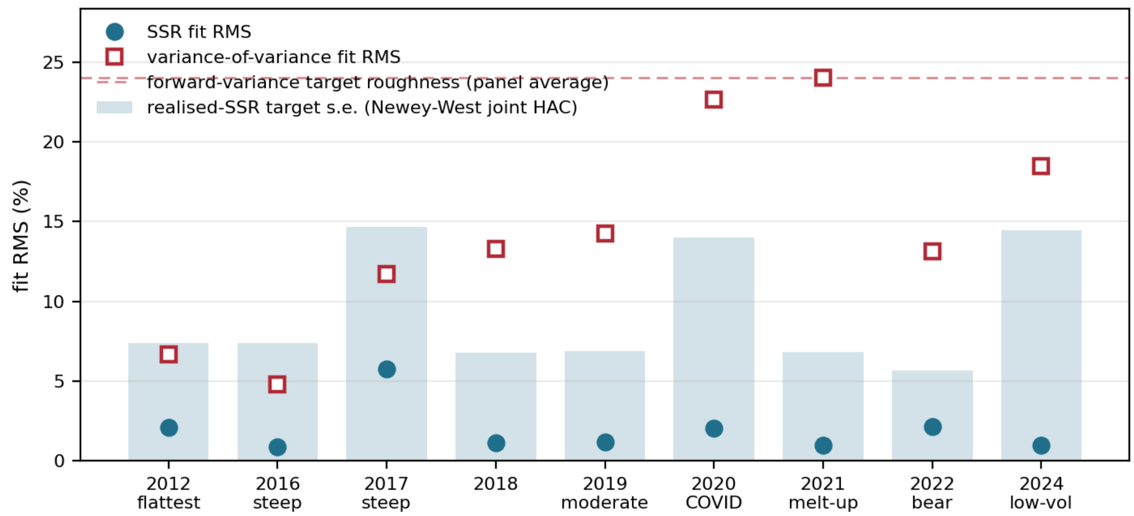

Figure 5: Off-SPX (NDX), both channels in one view, nine years. Filled circles: the realised-SSR fit RMS. Shaded bars: that year's displayed precision scale—the target's joint Newey–West HAC standard error, except at 2017, whose target is the Huber-robust β while the scale shown is OLS-derived (marked † in Table 11). Either way it is a scale to read the residual against, not a bound the residual could not fall below. Open squares: the variance-of-variance fit RMS, a realised target carrying no band of its own. Dashed line: that target's measured roughness—its RMS deviation from its own power law, 24%. That is how far the target departs from its own smooth shape—*roughness*, not a standard error and not a noise bound—so it carries none of the sampling interpretation the SSR band does, and the line is the *panel-average* roughness rather than each date's own. **Every SSR residual sits inside its band.**

5.6 What the calibrated kernel says about smile motion

Finding. The calibrated kernel places SPX on the sticky map in two independent coordinates: the smile's *level* rides with spot at the SSR, while its *shape* is very nearly frozen in moneyness.

Evidence. Taking the same exact-β readout across the whole at-the-money neighbourhood rather than at the level alone (Figure 6), the level responds as $\partial_{\ln S}\sigma_{\mathrm{ATM}} = \mathrm{SSR}\,\mathcal{S}$ with SSR running from 1.89 at one week down to 1.57 at three months—far from sticky-moneyness (SSR=0), between sticky-strike (1) and the local-volatility limit (2), and on the Doeff–Kamal $H+\frac{3}{2}$ backbone through the belly. The skew's own response $\partial_{\ln S}\mathcal{S}$, normalised by the rate at which a strike-pinned smile would swing it (twice the curvature), runs from 0.001 at one week to 0.03 at three months: the shape is essentially frozen in moneyness, with a small and realistic steepening on sell-offs that grows mildly with tenor. The one-month smile shifts up nearly in parallel on a 1% drop, unlike the flat sticky-moneyness or the tilted sticky-strike responses.

Interpretation. The textbook equity picture comes out as a result rather than an assumption: the fitted kernel holds the shape sticky and lets only the level ride, so the smile's whole response to a spot move collapses to the single number (39) folds into a delta.

Boundary. The frozen shape is a structural constraint as much as an empirical match: under the translation invariance of Section 3 the conditional smile is a function of the volatility regime alone, so shape responds to spot only through the leverage-induced regime shift. A *deterministic, spot-level-dependent* shape lies outside the model's reach.

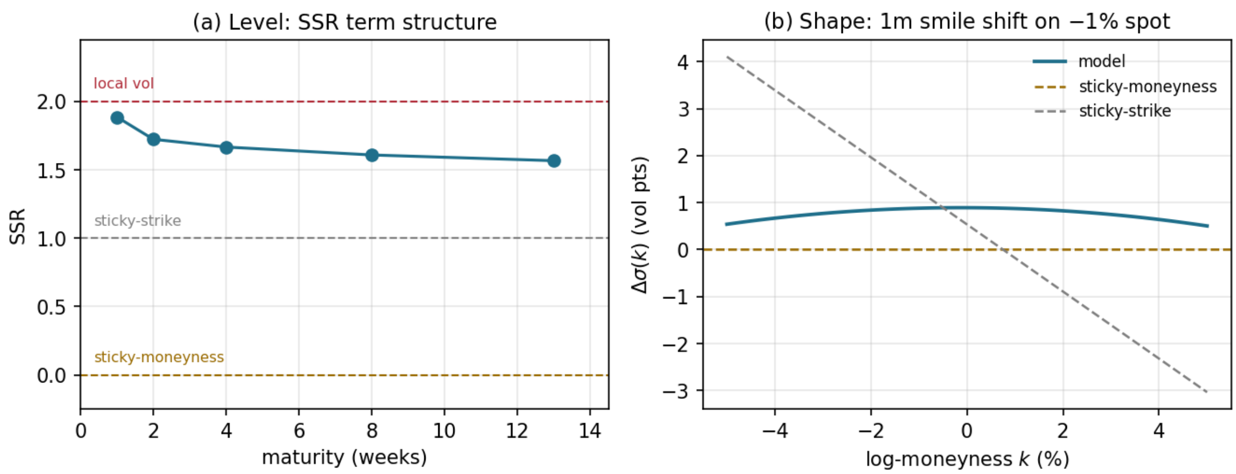

Figure 6: Where the calibrated model sits on the sticky map (SPX, conditional-smile response to a spot move, by moment). **(a)** *Level*: the SSR term structure, running 1.89 at one week to 1.57 at three months, between the sticky-strike and local-volatility reference lines. **(b)** *Shape*: the one-month smile's response to a -1% move, against the sticky-moneyness and sticky-strike references. The level rides and the shape does not.

6 Synthesis, limits, and conclusion

6.1 What is identified

Fixing the marginals does not fix the dynamics. With five smile observables held to a few basis points, the long-end skew-stickiness still moves across a band of width 0.05 around the calibrated fit: the gap European prices leave open is small enough to be a calibration target and large enough to matter.

Fitting the two calibrated blocks narrows the kernel: across the nine-regime panel the SSR residual RMS lies below the tenor-RMS estimated standard error at every date, and the forward-variance residuals—whose target carries no sampling band—are reported separately (Section 5.3). But not uniquely: distinct parameter vectors reproduce the same readouts along the soft directions Table 12 locates, median effective rank five of seven.

What the procedure identifies is therefore the tolerance set (28), a neighbourhood of the exact class (27), with the block-specific tolerances of (28). The parameter vector is loose—two accepted fits can differ by about a point of skew-stickiness RMS—while the readout it produces is stable under both perturbations that matter, the static date it is calibrated on and the sample it is used out of. The readout is what is carried downstream.

Read as a contract, that states what may and may not be claimed. Only outputs that are themselves functionals of the selected readouts inherit the identification; other prices are model-implied. The distinction bites precisely where it is tempting to ignore it: a forward-starting or path-dependent structure is identified in this sense only to the extent that its value depends on the joint law through the first-order spot–vol response and the forward-variance amplitude, and such prices can also depend on joint tails and on higher-order features $\mathcal{R}$ never sees. Two failures of identification then have different remedies. *Within* the readout set are directions $\mathcal{R}$ leaves soft, which Table 12 locates and which sharper targets of the same kind would tighten. *Outside* it, statistics no chosen readout sees are not weakly identified but unconstrained: the Zumbach effect and the broader time-reversal asymmetry of volatility, any asymmetry in the vol-of-vol itself, and the joint tail behaviour of spot and variance—[30] shows the question is live by testing the Zumbach effect explicitly.

6.2 What the two-timescale realization captures

Classical stochastic-local volatility also matches vanilla marginals and also produces smile dynamics; the distinction is which quantities are *independently imposed*. There the residual spot–vol response falls to the backbone, which is why calibrated backbones are documented to cluster [24]; here the overlay carries the marginal level, leaving the innovation leverages and the two persistence loadings free to be identified against an observed term structure.

Four features make the identification computable: the transition law is *explicit*, so prices, Greeks and readouts are deterministic fixed-resolution evaluations rather than particle [14] or grid computations; the martingale property is enforced *structurally*, by the branch law, not by penalty; the fast/slow split gives the spot–vol response a term structure with two interpretable timescales rather than a single decay; and return–regime dependence runs through a shared branch index, coupling leverage and variance dynamics inside the kernel rather than afterwards. Propagation stays in the finite-mixture class throughout, which is what keeps the whole chain deterministic. Against frontier stochastic-local volatility—multi-factor, rough, or neural—the edge is this combination, not dynamic control as such.

6.3 Limits revealed by the evidence

Four limitations survive the empirical study, each exposed by specific evidence and each pointing at a different kind of extension (Table 10). The short end is the weak part of the fit, and the held-out test of Section 5.4 measures how weak: withhold the one-week tenor and the model undershoots it by a median 11%, at seven of nine dates, while the withheld three-month tenor is recovered to 3%. The model skew-stickiness also rises to the $H{=}\frac{1}{2}$ ceiling 2 where SPX prints $\approx$1.5 sub-month; the conditional smile shape is near-frozen in moneyness, responding to spot only through the regime channel; in the off-index steep-decay years the realised variance-of-variance falls from 30 to 90 days at a ratio $\approx$0.4 against the kernel's floor $\approx$0.58; and observationally equivalent parameter vectors populate the soft Jacobian directions. None is a numerical defect.

One thing the evidence does *not* establish is a hedging edge. What is measured is a *smile-roll residual variance*: the mean square of $\Delta\sigma_{\mathrm{ATM}} - R\,s\,r$ over each window—the part of the at-the-money move that a one-parameter delta family $\Delta_{\mathrm{BS}} + \mathrm{Vega}\,R\,\mathcal{S}/S$ leaves behind—with s the strike slope and r the return. It is a proxy for hedging error, not a traded profit and loss: no vega

Limitation	Natural extension
Diffusive short end	rough or multiscale kernel (SSR $\to H+\frac{3}{2}$) [28]
Spot-level-independent conditional shape	state-dependent mixture weights
Two-timescale decay floor	richer persistence spectrum
Weak parameter identification	forward-start skew and variance-index smile data

Table 10: Limitations as a research map. The first two are consequences of the construction—the translation invariance that makes the readouts exact closed forms is what freezes the conditional shape—the third is a property of the two-factor family, and the fourth of the data available to identify it.

weighting, no position, no transaction cost. On that proxy the family is worth a great deal against no adjustment—R=0 carries 2.8–3.4× the residual variance of the industry minimum-variance delta—and nothing against a competent choice of R. Out of sample on 2015–2023, under the same train-only affine procedure, fitted separately to each competitor, and a 20-row purge between train and test—the map is fitted on 21-day forward outcomes, so without an embargo the last training rows would look into the first test origins—the calibrated kernel's own one-month readout, the corrected constant leg (the R=1.5 input mapped to the training-sample target mean), an analytic smile-implied R and a trailing at-the-money regression are all within 5% of the cross-strike Hull–White delta of [40] and of each other. The kernel leg is served under a strict lag: each annual fit is calibrated to that calendar year's realised SSR, so on any date in year Y the forecaster uses the most recent fit from a year strictly before Y, never the contemporaneous one. The minimum-variance delta collapses to estimating one number, and one number is estimable many ways, so the model's dynamic content has nowhere to show up. Where it should show up instead is in quantities the marginals cannot price at all—forward-start and path-dependent structures, whose very definition needs the forward implied volatility the marginals leave open [18], and for which the construction supplies deterministic finite-sum forward densities. That comparison is left to future work.

Two limitations deserve a word. The value 2 is the short-time limit of the skew-stickiness under a diffusion, not a bound on the finite-step statistic: where the realised one-week value exceeds it—as it does in the lowest-vol years, at ≈2.1–2.5—the kernel follows it there. And the off-index residual combines the two-factor decay floor with the physical measure in which that target must be quoted; the two cannot be separated with the data available there.

6.4 What would sharpen the identification

What would sharpen the identification is data, not a larger parameter count. The variance-index *smile*—its skew and tails, not only the at-the-money level—reads the vol-of-vol shape under $\mathbb{Q}$, and the atomic variance-index law of Section 4.4 prices it across strikes with no new machinery; the forward-start smile skew would supersede the realised proxy used here as the clean risk-neutral observable of spot–vol leverage, but it needs cliquet quotes absent from listed feeds. The framework may also be adapted to intraday surfaces, but jumps, seasonal liquidity, pinning and synchronized quote construction require a separate model and dataset [31].

Two directions of extension follow from the architecture rather than from this instance of it: the contribution is the architecture, not the two-factor kernel that occupies the framework here.

Refining the topology. Enlarging $\mathcal{R}_{\mathrm{cal}}$ shrinks both (27) and (28), so a new observable buys identification rather than fit. It buys it only under two conditions, neither automatic. It must be expressible as a functional of the discrete kernel—the readouts used here are closed forms because they are low-order functionals of a Gaussian-mixture law propagated by the composition of Section 3.4, and a lagged cross-moment or a tail functional must be shown to be one. And it must carry identifying power for the coordinates it is meant to pin, with the counting condition of Section 5.2 still holding after whatever parameters it brings with it. The binding constraint is observables, not flexibility.

Changing the occupant. The Gaussian-mixture kernel is a carrier, not a model. The two-timescale latent-regime specialisation of Section 3.2 is one process placed inside it, chosen for span and tractability rather than for economic commitment; the martingale lock, the leverage overlay that carries the marginals, the recompression that holds the component budget flat, and the deterministic finite-sum readouts are properties of the carrier. Any transition law expressible as—or approximable by—a Gaussian mixture can be carried in the same machinery, which includes Markovian approximations of rough kernels, further factors, and mixture components representing jumps. The cost is not architectural but dimensional: the carried state and the recompression budget grow, and the identification burden grows with the parameter count. The claims made in this paper are about the architecture; the two-factor occupant is what the available observables can currently support.

6.5 Conclusion

European option prices identify marginals and leave the transition kernel open. This paper takes that kernel as the unknown of an inverse calibration problem: the marginal layer is fixed independently by an arbitrage-free engine, an explicit finite Gaussian-mixture family supplies the candidates, and a representative of that family is calibrated to the specified skew-stickiness and forward-variance readouts—up to readout equivalence—rather than fixed by a pre-selected process. What the implementation delivers is weaker than the ideal admissible kernel it targets: martingality at the carried numerical state before recompression, and marginal compatibility measured through finite-feature discrepancies rather than imposed.

The evidence is that this works, and where it stops. The kernel fits nine SPX regimes to within the target's own sampling band; the skew-stickiness channel transfers to a second index on that index's own strip and realised increments, with the observation-operator correction those targets need estimated on SPX and transported; and the residual—a shape gap in steep-decay off-index years—is reproducible and bounded, and is consistent with a combination of the two-timescale decay and the measure in which that target is quoted that the available data cannot separate.

The broader implication is narrower than "dynamics can be calibrated from observed behaviour", which market models of implied volatility have done for two decades [32, 33, 34]. It is that the *transition kernel* between fixed, convex-ordered marginals can be selected that way—in place of a transport criterion, an entropic reference, or a prescribed stochastic backbone—and that the selection can be carried out in a family explicit enough for the selecting quantities to be deterministic finite evaluations of its own parameters, so that static no-arbitrage is a property of the construction rather than a constraint the dynamics must be made to respect. The remaining limitations point toward richer kernel families and fuller surface-level observations rather than a return to process-first selection.

Acknowledgements

I would like to thank my fellow traders Yan Guo, Jun Liu, and Ge Yi for several conversations during the preparation of this work.

A The SANOS marginal-calibration LP

For completeness, Algorithm 1 states the SANOS static-layer calibration in this paper's notation; the original is [1]. It fits, per maturity, a weights-only Gaussian mixture whose call surface lands within bid–ask and is arbitrage-free *conditional on* the continuum convex-order guarantee supplied externally by the marginal layer—the program displayed below enforces the calendar inequalities on a strike grid, not on the continuum—and free of butterfly and calendar arbitrage (positivity → valid density; forward constraint → martingale; convex-order inequality → calendar). The problem is convex—a linear program for a piecewise-linear smoothing penalty $\mathcal{R}$, quadratic for a quadratic one. Feasibility is a hypothesis on the quotes, not a property of the construction: when the bid–ask bands admit a jointly arbitrage-free surface the fitted solution stays inside them, and when they do not—mutually inconsistent quotes across strikes or maturities are not excluded a priori—the program as written is infeasible and slack variables or soft quote penalties are required. This is the admissible-marginal layer $\{\mu_j\}$ on which Theorem 1 and its convex-order precondition rest.

Algorithm 1 SANOS marginal-calibration LP at maturity T_j (this paper's notation; after [1]).

Require: strikes $\{K_{j,m}\}$ with bid/ask $[B_{j,m}, A_{j,m}]$, mids $C^{\text{mid}}_{j,m}$, weights $\omega_{j,m}$; forward F_j; fixed anchors $\{a_i\}_{i=1}^N$ in $z = \log(S/F_j)$; bandwidth h; previous-maturity fit q_{j-1}

Ensure: weights $q_j \geq 0$ with $\rho_{S,T_j}(z) = \sum_i q_j^i \, \mathcal{N}(z; a_i, h^2)$, arbitrage-free on the enforced grid

1: Build the linear pricing map $(\Pi_j)_{m,i} = \mathrm{BS}\big(F_j e^{a_i + h^2/2},\, K_{j,m},\, h\big)$, the price of basis component i at strike $K_{j,m}$.

2: Solve the linear program over $(q_j, t) \geq 0$:

$$\min \ \sum_m \omega_{j,m}\, t_m \ + \ \lambda_h\, \mathcal{R}(q_j)$$

$$\begin{aligned}
\text{s.t.}\quad & -t_m \leq (\Pi_j q_j)_m - C^{\text{mid}}_{j,m} \leq t_m && (L^1 \text{ fit to mids})\\
& B_{j,m} \leq (\Pi_j q_j)_m \leq A_{j,m} && (\text{hard bid–ask box})\\
& \mathbf{1}^\top q_j = 1, \quad \textstyle\sum_i q_j^i\, e^{a_i + h^2/2} = 1 && (\text{normalisation; forward})\\
& \widehat{C}_j(K_g) \geq \widehat{C}_{j-1}(K_g) \ \ \forall K_g \text{ on a dense grid} && (\text{calendar / convex order})
\end{aligned}$$

3: **return** q_j; the CDF is $G_{S,T_j}(K) = \sum_i q_j^i\, \Phi\big((\log(K/F_j) - a_i)/h\big)$.

Two qualifications on the restatement. First, the convex-order condition appears above as a family of call inequalities at grid strikes K_g. As written that is *numerical* enforcement— exact at the grid points, and beyond them only as far as the smoothness of the mixture basis carries it—rather than an exact finite reduction of the continuum constraint. Second, this is a restatement in the present notation and not a transcription: the authoritative formulation, and the precise sense in which its output is strictly arbitrage-free, are [1], and where the two differ the original governs. Theorem 1 takes the convex order of the delivered marginals as a *hypothesis*; this paper does not re-verify it on the fitted surfaces.

B Proofs of the approximation and stability results

Sections 2–4 use two mathematical facts, stated there as Propositions 2 and 3. This appendix proves them, in the order in which each is used:

$$\text{leverage and kernel stability} \implies \text{dynamic-readout continuity.}$$

Both concern the construction of Section 3 as it is built: the overlaid finite kernel, and the readout evaluated on it.

Notation. We keep the main text's objects throughout: the lifted state $X_j = (Z_j, U_j)$ with $U_j = (u_j^{\mathsf{F}}, u_j^{\mathsf{S}}) \in \mathbb{R}^2$ continuous, its law $\widehat{\mu}_j(\mathrm{d}z, \mathrm{d}u)$, and the admissible kernels $\mathcal{K}_j \in \mathcal{A}_j$ of Section 2.2, equation (5). In gross-return coordinates $G_{j+1} = e^{Z_{j+1}-Z_j}$ the martingale constraint (19) is affine, $\mathbb{E}[G_{j+1} \mid X_j] = 1$, and a Gaussian log-return component is exactly a lognormal gross-return component, so the two coordinate systems describe the same family.

Distances. On laws of the lifted state, W_1 is the 1-Wasserstein distance under the metric $d\big((z,u),(\tilde{z},\tilde{u})\big) = |z-\tilde{z}| + \|u-\tilde{u}\|_2$. On kernels we use the integrated conditional distance

$$W_{1,\mu}^{\mathrm{ker}}(\mathcal{K}, \widetilde{\mathcal{K}}) = \int W_1\big(\mathcal{K}(x,\cdot), \widetilde{\mathcal{K}}(x,\cdot)\big)\, \mu(\mathrm{d}x), \tag{44}$$

the source law μ being the lifted law the kernel is applied to. Coefficient vectors carry the Euclidean norm $\|\cdot\|$ on $\mathbb{R}^{n_\theta}$, and leverage functions the norm $\|\ell\|_{L^2(\mu)}^2 = \int \ell(z)^2\, \mu(\mathrm{d}z)$ against the spot marginal of the same μ.

B.1 Martingality: the ideal statement and the implemented one

Two facts are used throughout and are separated because they condition on different σ-algebras. Both are immediate from the coefficient map.

Proposition 4 (Ideal kernel, fibrewise)**.** *For the intrinsic kernel with drift $d_\ell(u) = \widetilde{m}_\ell(u) - A(u)$ of* (19)*, $\mathbb{E}\big[e^{Z_{j+1}} \mid Z_j = z,\, U_j = u\big] = e^z$ for every (z,u) and every parameter value.*

Proposition 5 (Implemented kernel, at its numerical state)**.** *Let $\mathcal{G}_j$ be generated by the component index and the price abscissa the propagation carries. For the overlaid kernel with the lock* (23)*, and* before *the recompression projection, $\mathbb{E}\big[e^{Z_{j+1}} \mid \mathcal{G}_j\big] = e^{Z_j}$ identically in the coefficients and in L. The projection that follows preserves mass and the unconditional forward but not this conditional identity, so no martingale property is claimed for the reduced object.*

Proof. Conditionally on a branch the increment is Gaussian, so $\mathbb{E}[e^{\Delta Z} \mid \cdot, \ell] = \exp(d_\ell + \frac{1}{2}V_\ell)$. Averaging over the branches with weights w_ℓ and inserting $A(u)$ gives 1; averaging instead over the joint quadrature (k,ℓ) with weights $\varpi_k w_\ell$ and inserting $A_c^{\mathrm{LV}}(z)$ gives the second statement. □

Proposition 5 is the weaker of the two and is the one the production chain supports; every use of "martingality is exact" in the main text refers to it.

B.2 Standing hypotheses

All statements below are local, on a compact region of the implementation, and at a *fixed* numerical pattern.

Assumption 1 (Nondegenerate region, fixed pattern)**.** (i) Coefficients. *$\boldsymbol{\theta}$ lies in a compact set with $|\kappa_{\mathsf{F}}|, |\kappa_{\mathsf{S}}| \le \kappa_{\max} < 1$; the log-variance clamp of Table 13 bounds V_ℓ into $[V_{\min}, V_{\max}]$, $V_{\min} > 0$, uniformly in the node count.* (ii) Local variance. *The regularised input satisfies $0 < a_{\min} \le a \le a_{\max}$ and the spot-conditional variance $m(z) = \mathbb{E}[\nu(U) \mid Z{=}z] \ge m_{\min} > 0$; σ_{Dupire} is supplied by the marginal engine and its convergence is external to this paper.* (iii) Readout. *Strikes are bounded away from the wings; at-the-money Black vega is bounded below; the stationary average skew and the return variance in* (37) *are bounded away from zero.* (iv) Conditioning map. *The spot-conditional variance is stable in the source law: there is C_m with $\|m_{\widehat{\mu}} - m_{\widehat{\nu}}\|_{L^2(\mu^Z)} \le C_m\, \mathrm{d}_{\mathrm{ad}}(\widehat{\mu}, \widehat{\nu})$ for the adapted distance d_{ad} in use. This is assumed, not derived: conditional expectations are not Lipschitz in weak or adapted topologies without further structure, and we do not establish that* *structure here.* (v) Source regularity and pattern. *The source kernel is Lipschitz in its state, $W_1\big(\mathcal{K}_j(x,\cdot),\mathcal{K}_j(\tilde{x},\cdot)\big) \le L_j\, d(x,\tilde{x})$ with $L_j < \infty$ on the region—which is what lets one-step gaps be propagated along the maturity grid. The branch and factor quadratures, the collocation rule and the recompression partition are held fixed. Perturbations are in $\boldsymbol{\theta}$, in the local-variance input and in the source law—never in the pattern—so nothing below is a statement about refining the recompression, and the paper makes no such claim (Section 2.3).*

Part (i) is a hypothesis on the *realised* construction rather than a consequence of it. Two features make it stable under refinement: the persistent factors are not discretised, so refining the remaining quadrature refines an integral against a fixed Gaussian rather than enlarging the state space; and the log-variance is clamped before exponentiation, so V_ℓ is bounded uniformly in the node count rather than only at each fixed one. The resolution-agreement check of Section 5.2 is retained as a diagnostic.

B.3 Stability of the leverage pipeline

Proposition 6 (Local stability at a fixed numerical pattern)**.** *Fix the numerical pattern of Assumption 1(v). On the nondegenerate region, and given the conditioning hypothesis (iv), the maps $(a,m) \mapsto \ell^2 = a/m$ and $(\theta,\ell) \mapsto \mathcal{K}_{\theta,\ell}$ are locally Lipschitz—the second in the integrated conditional Wasserstein distance (44), with $W^{\mathrm{ker}}_{1,\mu}(\mathcal{K}_{\theta,\ell},\mathcal{K}_{\tilde{\theta},\tilde{\ell}}) \le C_\theta\|\theta-\tilde{\theta}\|+C_\ell\|\ell-\tilde{\ell}\|_{L^2(\mu)}$—and composing them along a finite maturity grid gives $e_{j+1} \le \varepsilon_j + L_j e_j$, where $e_j = W_1(\mu_{j,n},\mu_j)$ is the propagated-law gap and $\varepsilon_j = \int W_1(\mathcal{K}_{j,n}(x,\cdot),\mathcal{K}_j(x,\cdot))\,\mu_{j,n}(\mathrm{d}x)$ the one-step kernel gap.*

Proof. The first is the quotient rule with $m \ge m_{\min}$ and $a \le a_{\max}$, constant $a_{\max}/m_{\min}^2 + 1/m_{\min}$; ν is bounded under the clamp. For the second, each branch mean and variance is smooth in (θ,ℓ) on the compact region and a Gaussian mixture's W_1 is bounded by the weighted sum of componentwise mean and standard-deviation gaps; C_θ, C_ℓ collect those bounds over the fixed node set. The recursion follows by inserting $\mu_{j,n}K_j$ and using the Lipschitz property of K_j in its source. □

Four limitations are part of the statement. The stability of the conditioning map is a hypothesis, not a result—Assumption 1(iv)—so the chain of implications starts one step later than it appears to. It is local, on a region assumed rather than derived. It holds at one fixed pattern: it says nothing about refining the recompression partition, and in particular does not assert that the reduced chain converges to the unreduced one. And it concerns the overlaid kernel, not the full production pipeline—the recompression projection of Section 3.4 is not a kernel and is not covered here; its effect is measured, not bounded. Proposition 2 in the main text is this result applied to a convergent sequence of inputs at a fixed pattern.

B.4 Readout continuity

Theorem 7 (Readout regularity)**.** *Under Assumption 1, each block of $\mathcal{R} = (\mathcal{D},\mathcal{S},\mathcal{V})$ is locally Lipschitz in the mixture state and in $\boldsymbol{\theta}$, and $\mathcal{R}$ is therefore locally Lipschitz and piecewise C^1 in $\boldsymbol{\theta}$; the pieces are the segment boundaries of the calendar interpolation.*

Proof. $\mathcal{D}$ is a finite sum of normal CDFs with σ bounded below, and Φ is globally Lipschitz; bare W_1 would not suffice here, and the result uses the implementation's explicit smooth mixture CDF. $\mathcal{V}$ composes the closed-form k-step moments (40), a Gauss–Hermite sum of exponentials of affine functions under the clamp, a square root away from zero, and the at-the-money Black inverse with vega bounded below; the expiry variances satisfy $V_k^{\mathsf{FF}}, V_k^{\mathsf{SS}} \ge 1-\kappa_{\max}^2 > 0$ for $k \ge 1$, so the node map is Lipschitz and no separate lower bound on forward variance is assumed. $\mathcal{S}$ is a finite sum of bounded factors over a return variance and a stationary average skew, both bounded away from zero by Assumption 1(iii), with the trilinear interpolant of (36) Lipschitz with the same constant as its data. □

C The realised-SSR sampling band (HAC)

The realised SSR markers and the shaded band in Figures 3 (SPX) and 5 (NDX) are a physical-measure ($\mathbb{P}$) estimate from a single year of daily end-of-day chains, so both the numerator and the denominator carry sampling error that the band makes explicit. Comparing this $\mathbb{P}$ target to the kernel's risk-neutral readout is motivated, at leading order, by the fact that the SSR is a ratio of one-step *covariations*: quadratic covariation is measure-invariant—the $\mathbb{P}/\mathbb{Q}$ drift enters at $O(\mathrm{d}t)$ and drops out of the leading covariance—so the *within-state* comovement channel reads the same under either measure to that order. Two limits on that argument should be kept in view. It does not extend to the between-state terms of the pooled annual regression, which can carry physical drift and risk-premium variation that the risk-neutral kernel does not represent (Section 4 measures their size). And the band below quantifies sampling uncertainty of the empirical pooled statistic; it is not evidence that the empirical and model functionals coincide. The higher-moment vol-of-vol is *not* so protected (Section 5.2). At each tenor T_j ($j = 1, \ldots, 5$; 1w to 3m) we regress the daily ATM-forward implied-vol change $\Delta\sigma^*_{j,t}$ on the daily log return r_t with an intercept, and divide the slope by the mean daily ATM skew $\bar{\mathcal{S}}_j = \overline{\partial_{\ln K}\sigma}\,|_{\mathrm{ATM}}$ at T_j:

$$\Delta\sigma^*_{j,t} = a_j + \beta_j\, r_t + \varepsilon_{j,t}, \qquad \widehat{R}_j \;=\; \beta_j/\bar{\mathcal{S}}_j. \tag{45}$$

The ratio is Bergomi's [22, 23], and as an instantaneous ratio of covariations it carries no estimation window: the window is a choice, not part of the definition. The calendar-year block used here follows from the calibration design rather than from precedent—one kernel is fitted per regime, so one target is required per regime. Published series may use much shorter windows, sixty days in [30] for instance, which suits displaying the ratio as a time series; at that length the sampling band derived below would exceed the fit errors of Section 5.3 several times over and so could not adjudicate them.

Both pieces are estimated on the same days, so $\widehat{R}_j$ is a ratio of two *jointly* estimated quantities and its standard error is not the combination of two separate ones. Treat $(\hat\beta_j, \bar{\mathcal{S}}_j)$ as one M-estimator, with per-day influence functions

$$\psi_{j,t} = \Big(\big[(X^\top X/n)^{-1} X_t\, \hat\varepsilon_{j,t}\big]_\beta,\;\; \mathcal{S}_{j,t} - \bar{\mathcal{S}}_j\Big)^\top, \tag{46}$$

where X is the design matrix, and estimate their joint long-run covariance by a Bartlett-kernel HAC, which guarantees a non-negative variance:

$$\widehat\Omega_j = \widehat\Gamma_{j,0} + \sum_{\ell=1}^{L}\Big(1 - \tfrac{\ell}{L+1}\Big)\big(\widehat\Gamma_{j,\ell} + \widehat\Gamma_{j,\ell}^\top\big), \quad \widehat\Gamma_{j,\ell} = \tfrac{1}{n}\textstyle\sum_t \psi_{j,t}\,\psi_{j,t-\ell}^\top, \quad L = \big\lfloor 4\,(n/100)^{2/9}\big\rfloor, \tag{47}$$

with the automatic lag $L \approx$ 4–5, so that $\widehat{\mathrm{Var}}(\hat\beta_j, \bar{\mathcal{S}}_j) = \widehat\Omega_j/n$. The delta method applied to $g(\beta, \mathcal{S}) = \beta/\mathcal{S}$ then gives

$$\mathrm{se}(\widehat{R}_j) = \sqrt{\nabla g^\top\,(\widehat\Omega_j/n)\,\nabla g}, \qquad \nabla g = \big(1/\bar{\mathcal{S}}_j,\; -\hat\beta_j/\bar{\mathcal{S}}_j^2\big)^\top, \tag{48}$$

and the plotted strip is $\widehat{R}_j \pm \mathrm{se}(\widehat{R}_j)$, one standard error per tenor.

Two features of (48) are worth isolating, because the more familiar shortcut—a HAC slope error combined with $\mathrm{std}(\mathcal{S}_{j,\cdot})/\sqrt{n}$ as though the two were independent—gets both wrong, and in opposite directions. The daily ATM skew is strongly persistent within a year, so an iid standard error understates it: the corresponding diagonal of $\widehat{\Omega}_j$ inflates $\mathrm{se}(\bar{\mathcal{S}}_j)$ by a factor with median 2.1 (range 1.2–2.2), remarkably stable across tenors and years. Against that, $\hat{\beta}_j$ and $\bar{\mathcal{S}}_j$ are strongly *positively* correlated—HAC correlation +0.46 at the median, interquartile range +0.37 to +0.54, positive in 89 of the 90 index-year-tenors—because a year with a steeper mean skew is a year with a steeper vol–return slope. Both are negative quantities, so the two entries of ∇g carry opposite signs and that common variation largely *cancels* in the ratio: the cross term in (48) enters negatively, which no independent combination can represent. This second effect is the larger, so the shortcut is not conservative in the way it looks—it *overstates* $\mathrm{se}(\widehat{R}_j)$, by a median 9% across the eighteen index-years reported here, and by as much as 39%. The scalar *band* quoted in each panel title is this standard error as a single percentage—the tenor-RMS of the relative error,

$$\text{band} \;=\; 100 \times \sqrt{\tfrac{1}{5} \textstyle\sum_j \left(\mathrm{se}(\widehat{R}_j)/\widehat{R}_j \right)^2}\ \%. \tag{49}$$

The band is a *reporting benchmark*, not part of the objective: the kernel is fit to the point estimates $\widehat{R}_j$ in equal-percentage least squares (29), and nothing in the optimiser consults the band—the fits stop on `xtol`. A fit whose SSR RMS falls below it sits, on average, inside the ±1-se strip; it is *not* the case that the residual cannot be smaller, since the kernel carries more free parameters than the SSR block has tenors and can interpolate the point estimates exactly. Nor is this a multivariate test: that would need the cross-tenor covariance of $\widehat{R}$ together with positive residual degrees of freedom, and the second is unavailable on this block whatever is done about the first. What the band fixes is a resolution below which further reduction is not interpretable, so we stop there rather than chase sub-noise structure. The strip widens at the short end, where the slope regression has the fewest effective observations and the thinnest short-dated options. The forward-variance/VIX ATM implied volatility readout (SPX, Figure 4) carries no band: it is a single-day risk-neutral ($\mathbb{Q}$) quote, not a sample statistic.

A nonparametric check. Equation (48) is an asymptotic formula applied at $n \approx 250$, so we confirm it without the asymptotics. A moving-block bootstrap resamples contiguous runs of trading days—block lengths 5, 10 and 20, 2000 replicates—and recomputes the *entire* statistic on each replicate: slope, skew mean and their ratio, with the blocks drawn on common day indices across the five tenors so that the cross-tenor dependence the RMS band depends on is preserved. Serial dependence is then carried by the blocks rather than modelled, and no linearisation is involved. The resulting bands agree with (48) to a median ratio of 1.02 (range 0.89–1.14), while both sit below the independent shortcut; the block length moves them by less than that gap—two constructions this different agreeing is what licenses (48) at $n \approx 250$. No acceptance call in Section 5 depends on the choice: every reported fit sits inside its band under all three constructions.

D Data construction and filters

Both realised targets are estimated on ORATS end-of-day chains. Neither is usable raw. This appendix records every filter, the threshold it uses, and the evidence that it is inert where the data are already clean; nothing here is tuned per year.

The realised skew-stickiness target. The realised SSR is a $\mathbb{P}$ regression of daily ATM-vol changes on returns (Appendix C), so a single corrupt print contaminates a whole year. Three filters precede it.

(i) Day spot. The day's spot is the *median* `stockPrice` across the day's smiles. ORATS records this field per option row and it is unreliable—it varies across strikes and is corrupt by up to +20% in 2022–23—so taking any single row injects spurious ±20% one-day returns.

(ii) Positive fitted ATM skew. An index smile is always negative-skew, so a maturity whose *fitted* ATM skew is positive on a given day is a corrupt short-dated quadratic fit through thin options, and that maturity is dropped for that day.

(iii) Isolated short-end volatility spikes. Thin weeklies—pronounced on NDX—occasionally print an absurd short-dated ATM vol (20–45%, up to 2.5× the one-month level, on *calm* days). These survive (ii), their skew being negative. We drop a short maturity whose ATM vol exceeds 1.5× the one-month: a genuine crash lifts the whole curve and keeps the 1wk/1m ratio below 1.5, whereas the artifact is an isolated short-end spike.

Rules (ii)–(iii) are *inert on clean data.* On liquid SPX they remove essentially nothing— bands and fits are unchanged and the COVID crashes are retained—while on NDX 2012/2016/2017 rule (iii) removes 5/24/35 corrupt days, restoring the one-week SSR from 0.75–1.02 to ≈1.4–1.6 and its target standard error from 20–41% to 8–15%. NDX 2017 is the one year where a large anti-leverage one-week print survives all three filters. Its effect is confined to the shortest tenor and is large there: the ordinary least-squares β returns 1.42 at one week against 1.77 under a Huber-robust estimator, and the two agree to within 3% at every other tenor. The OLS estimate also inverts the term structure, placing the one-week value *below* the two-week one, which no index skew-stickiness curve does. For that year, and only that year, the target is the Huber-robust β, marked † in Table 11, where the scale shown beside it is OLS-derived rather than that target's own standard error.

Scheduled events are in scope by construction. Removing days adjacent to scheduled macro releases shifts the estimated SSR term structure by a median 9% (max 15% across 2015–2024), almost entirely through β—the mean skew moves by under 1.5%. The target therefore embeds scheduled-event risk, which is the intended scope: the kernel is fit to the dynamics the market actually prices, not to a release-free counterfactual.

The realised variance-of-variance target (NDX). NDX has no volatility-index options, and its variance-swap strip has only sparse monthly and quarterly expiries past the front, so a fixed-tenor series built from the *nearest* expiry inherits a spurious jump each time that expiry rolls. Each tenor is therefore built by the constant-maturity interpolation (50). Three further corrections are needed before the series is a usable target, each arrived at by measurement.

The estimator is the following. On trading day t let $\{(T_i, \hat{\sigma}_{t,i})\}_i$ be the model-free variance-swap volatilities at every listed NDX expiry with $T_i \geq 7$ days, each computed from that expiry's out-of-the-money strip by the standard log-contract replication. For a target tenor τ bracketed by $T_i \leq \tau \leq T_{i+1}$, interpolate *total* variance linearly in maturity and convert back to an annualised volatility,

$$V_t(\tau) = \Big[\tfrac{1}{\tau}\big((1-f)\,\hat{\sigma}^2_{t,i}\,T_i + f\,\hat{\sigma}^2_{t,i+1}\,T_{i+1}\big)\Big]^{1/2}, \qquad f = \frac{\tau - T_i}{T_{i+1} - T_i}, \tag{50}$$

with τ and the T_i in the same units, so that the day-count factor cancels and $\hat{\sigma}_{t,i}$ and $V_t(\tau)$ carry the same annualisation, so $V_t(\tau)$ is a *volatility* level in VIX units—not a variance and not a total variance—and is the exact analogue of the quantity $X = \sqrt{Y}$ whose at-the-money option the model prices in (43).

(i) A physical validity bound. Levels are restricted to $[0.02, 2.0]$ in annualised volatility. This is not a percentile but the range in which an index variance-swap level can exist at all. Of the 25,576 NDX observations across eleven tenors and ten years, *exactly one* lies outside it—a reading of 10^{-4}, one basis point of volatility—and its single increment carries 92.5% of that tenor-year's entire variance, driving the 2018 45-day estimate to 9.74 against neighbours near 1.5. For scale the distribution is $p_{0.01} = 0.084$, median 0.209, $p_{99.9} = 0.660$, maximum 0.835. A bound of this kind cannot remove signal: nothing trades at one basis point.

(ii) Adjacency. Days on which τ is not bracketed by two listed expiries are dropped rather than extrapolated, and log changes are formed only between days that are *adjacent* in the trading calendar. Letting a two-day change enter as if it were a one-day change is not innocuous at the thin tenors, where it moves the estimate by up to 59% (2021, 7 days); at tenors with full coverage the two agree to within a few tenths of a percent.

(iii) Bracket churn beyond three months. Constant-maturity interpolation removes the roll in the *target* maturity but not in the *bracketing pair*: occasionally the near expiry rolls past τ or a newly listed expiry slots in between, and the interpolation switches to a different pair of smiles. Identifying the pair by absolute expiry date—not by days to expiry, which counts down daily and would flag every day—such changes occur on 29.6% of days but carry 38.0% of the squared variation. The structure across tenors is the reason the correction is applied only beyond three months:

tenor	14d	21d	30d	45d	60d	90d	120d	180d
churn days (%)	59.2	58.6	52.2	33.4	19.1	5.7	8.7	**7.5**
$\lvert r\rvert$ churn / quiet	1.21	1.27	1.04	1.09	1.62	1.47	1.19	**2.11**
variance concentration	1.08	1.06	1.04	1.08	1.77	2.25	1.31	**4.63**

Where expiries are dense—weeklies at the front—a bracket switch is nearly seamless: frequent, but ordinary. Where they are sparse, the pair spans a wide gap and the switch is a genuine discontinuity: rare, but violent. Increments spanning a change are therefore dropped at 60 days and beyond and retained below; the 60-day tenor is recovered by this correction rather than excluded as an unlisted gap-centre.

The target is then, with $t_1 < \cdots < t_n$ the retained days of the calendar year and $r_k = \log V_{t_{k+1}}(\tau) - \log V_{t_k}(\tau)$,

$$\begin{aligned} &\xi(\tau) = \sqrt{252}\ \operatorname{sd}\big(\{\, r_k : \ k \text{ retained} \,\}\big), \qquad k \text{ retained when} \\ &\qquad t_{k+1} - t_k = 1 \text{ trading day}, \quad V_{t_k}, V_{t_{k+1}} \in [0.02, 2.0], \quad \text{and no bracket change if } \tau \geq 60\,\text{d}. \end{aligned} \tag{51}$$

Three conventions are choices and are recorded as such: log rather than percentage changes; demeaning rather than a zero-mean second moment; and $\sqrt{252}$ annualisation on trading days. The fitted tenors are 14, 21, 30, 45, 60, 90, 120 and 180 days. Seven days is excluded on coverage—it averages 26% of trading days and is a single day in three of the nine years—and 270/365 days on the absence of a bracketed volatility-index quote against which to correct the observation operator. From 14 days out coverage is at least 93%.

Two statistical screens were built and rejected, each by a diagnostic fixed before it ran. A rolling-median filter on the log level was rejected because it screened 2020 *hardest* (2.41% of points, cutting that year's 180-day estimate by 61.5%): its scale is the deviation from a lagging median, so a fast-moving year sets a tight band precisely when the market moves. A round-trip test—flagging a large increment immediately undone—was far more conservative (0.20%) but bought almost nothing on an independent criterion it was not designed for, namely that the term structure should decay with tenor: violations fell only from 15 to 14 of 63, and 2020 got worse. Both, moreover, *missed* the one observation the physical bound catches, the median filter because it re-centres on a corrupt neighbourhood and the round-trip test because that observation sits at the start of its series and has no increment into it to judge.

The observation operator, and correcting for it. The model side applies (43) at each fitted tenor, which is *not* the same functional as (51): one is the implied volatility of an at-the-money option on the terminal law of $\sqrt{Y}$ at horizon τ, whose variance window is fixed at thirty days; the other is the realised volatility of a time series of $V_t(\tau)$, whose window is τ itself. Both are annualised volatilities of a volatility level, which is what makes the comparison meaningful, but they are not counterparts, and the gap between them widens with τ because the realised object grows smoother as its own tenor lengthens while the readout stays pinned at thirty days.

SPX is the one underlying carrying both sides, so the gap is measurable rather than assumed. Pricing the VIX-implied $\mathbb{Q}$ at-the-money volatility against a realised $\mathbb{P}$ variance-of-variance built from SPX's own strip by the identical construction, the ratio is smooth and monotone in tenor—0.33 at seven days, 0.77 at thirty, crossing unity near three months, 1.22 at six—over nine to ten years of coverage at every tenor. Two effects of opposite sign are netted in it and this measurement does not separate them: a volatility-risk premium, which lifts the $\mathbb{Q}$ side, and the fact that vol-index options are written on *futures* whose volatility is damped by mean reversion relative to the spot the realised side measures, which lowers it. Neither is the largest term: the functional mismatch above is, and it is pure construction rather than measure. Applied per year and per tenor, the correction moves the demanded decay exponent from 0.44 to 0.18.

What can be checked, since the correction cannot be verified where it is applied, is whether it improves prediction on tenors the fit never saw: fitted at the 30- and 90-day anchors alone, the corrected-target run beats the raw-target run on the seven held-out tenors by a mean 32.5% at nine of nine dates under corrected scoring, against a 15.2% win at seven of nine for the raw-target run under raw scoring; absolute held-out RMS runs 31–46%. Each scoring favours its own training object, so what supports the correction is the gap between those margins. The churn correction was checked the same way and against a control: fitted at 14–45 days alone, it improves held-out 60–180 day prediction by 21.7%, while dropping the *same number* of increments chosen at random improves it by 0.3% over forty draws. The gain is attributable to those increments specifically and not to thinning a noisy series.

What the cleaning does to the conclusion. Much of the apparent steep-decay shape gap co-moves with the observation operator rather than with the kernel, and the supporting diagnostic is that SPX inherits the same apparent ceiling when fed realised targets. That is consistent with target construction contributing materially; it does not exclude kernel misspecification. What survives is stated in Section 5.5.

E Calibration protocol and robustness

One protocol is used for every reported fit. This appendix states it, quantifies the multi-basin freedom that makes an acceptance rule necessary, and reports the robustness experiments summarised in Section 5.3.

Objective. The objective is (34) at the design constants of Section 5.2, with no per-year tuning anywhere and one documented difference between the two underlyings: the skew-stickiness weights ς. On SPX they are unity, so the five tenors enter equally. On NDX they are the inverse of each tenor's own measured relative HAC standard error, normalised to $\overline{\varsigma^2} = 1$. That normalisation is what keeps the choice from being a second lever: the SSR block's expected squared magnitude is unchanged, so the SSR-versus-forward-variance balance is untouched and the weighting only redistributes *within* the block. The reason it is applied off index is that the off-index targets are far more unevenly determined—at NDX 2017 the one-week standard error is 24% of the target against 5.7% at three months, so equal weighting spends the fit on the least-determined cell. Realised weights run from 0.24 to 1.36 across the nine dates, their spread widest where the short-tenor errors are. One caveat travels with the choice: the recorded standard error is the ordinary least-squares estimator's, so at NDX 2017—the single date whose fitted target is the Huber-robust β—the weight and the target come from different estimators. The number reported beside that target is therefore an OLS-derived precision scale, not an uncertainty measure for the robust estimate; we have not computed a compatible robust one, and it should not be read as that date's target standard error.

At the five tenors 1wk–3m the SSR target $\mathcal{S}^\star$ is an ordinary least-squares β over a calendar-year window after the filters of Appendix D; the one exception, NDX 2017, is discussed there. The ridge is taken toward the current stage's anchor θ^0—the fixed uniform start in the cold stage, the cold stage's own optimum in the warm one—so the warm refinement is regularised toward where it begins rather than toward a fixed point of the parameter box.

Numerical resolution: one setting, and no ladder. The propagation does not discretise the volatility factors into abscissas the kernel walks: they are carried continuously and *integrated*, so refining the readout is an ordinary quadrature refinement of a fixed Gaussian rather than an enlargement of the state space. There is consequently no node ladder, and no node-count convergence question distinct from the quadrature limit.

Every fit reported here therefore runs at a single fixed resolution, and the readout's sensitivity to it is measured rather than assumed. Holding the fitted $\boldsymbol{\theta}$ and refining every quadrature the readout touches moves the forward-variance reading by at most 3.8%, monotonically decreasing and growing mildly with tenor; the SSR block does not pass through that path and does not move. Splitting the refinement shows roughly two-thirds of it is quadrature and one-third the recompression budget, the two being additive to within 0.1 percentage points. Several quadratures—in particular the stationary bivariate grid—leave the readout *bit-identical* at any setting, since the closed-form step law does not consult them.

The leverage overlay's finite-step remainder. The overlay matches the continuous relation (21) to leading order, and (24) names what is left over. Evaluated by `leverage_remainder.py` at the nodes the production step visits and with the production weights, δ_{LV} has a probability-weighted RMS of 0.40–3.19% across the nine shipped fits and a weighted median of 0.19–1.02%; the 99th weighted percentile runs 1.4–14.0%, worst at 2020.

Because δ_{LV} is a relative *variance* error the corresponding volatility error is about half of it: roughly 4 bp of implied volatility at RMS on a 20% surface at the best date and about 32 bp at the worst, against the 26 IV-bp marginal discrepancy the overlay exists to close (Table 8). At most dates the remainder is well inside that residual; at 2016 and 2020—the two steep-leverage years—it is of the same order, and at the worst date it exceeds it. We report it rather than repair it: correcting it would require an exact scalar solve for $L(z)$ at every price abscissa and a full recalibration. That is the honest statement of the trade—the remainder is a known, measured approximation of the same magnitude as the marginal discrepancy at the two steepest dates, not one demonstrably below it.

The resolution is chosen on cost. Each refinement step multiplies the readout cost by roughly six, so a fully refined reading is about 87 times the production one: the 58-minute panel below would take over three days. Against a $\leq 4\%$ readout bias on one of the two blocks, that is not a trade worth making for a model intended to be recalibrated daily, and the bias is reported instead. Its direction is known and favourable: the production reading *overstates* the forward-variance level slightly and more so at longer tenors, so the reported forward-variance residuals are, if anything, conservative—and the model's vol-of-vol term structure is marginally steeper than the reported figures imply.

Cost, per year. Every fit reported here was produced on one Apple-silicon GPU (MPS) at the single fixed resolution above, and the wall time is recorded in each fit record rather than estimated. Stage 1 (cold) runs 189–324 s per date and stage 2 (warm) 84–173 s, 0.56× stage 1; the protocol costs both at every date before the acceptance rule can compare them, 57.6 minutes across the nine SPX dates. Off index the eight-tenor NDX panel is a single cold stage, 39.4 minutes in total. Fits take 13–27 objective and 4–16 Jacobian evaluations, so a single-date recalibration is a few minutes—the operating point the design targets.

Determinism. The accelerated backend is bit-deterministic on device: repeated evaluation of the same $\boldsymbol{\theta}$ reproduces the residual and the Jacobian exactly. It matters most for the Jacobian—forward tangents pass through segment-boundary comparisons, where a boundary flip changes the derivative structure outright rather than perturbing its value—and it costs about 49% on a whole fit, leaving the accelerated path roughly thirteen times faster than the reference loop it replaces.

		SPX (forward variance from VIX)				NDX (realised strip, ς-weighted SSR)			
Date	regime	SSR	s.e.	vov	$d_{\mathcal{V}}$	SSR	s.e.	vov	$d_{\mathcal{V}}$
2012	flattest	2.02	7.0	8.75	6	2.10	7.4	6.66	8
2016	steep	1.74	5.1	4.68	9	0.88	7.4	4.81	8
2017	steep/calm	2.14	11.9	2.96	10	5.75	$14.7^{\dagger}$	11.72	8
2018	—	1.29	10.6	3.95	9	1.14	6.8	13.31	8
2019	moderate	0.86	5.2	5.82	9	1.18	6.9	14.28	8
2020	COVID	3.13	16.4	9.17	9	2.08	14.0	22.64	8
2021	melt-up	1.57	5.7	3.46	12	1.01	6.8	24.02	8
2022	bear grind	1.29	6.0	1.79	12	2.17	5.7	13.16	8
2024	low-vol	1.21	14.7	5.66	12	0.98	14.5	18.47	8
mean		**1.70**		5.14		**1.92**		14.34	

Table 11: Fit quality across regimes and underlyings, RMS in percent. **Every SSR entry lies below its displayed precision scale.** Except for the daggered row, that scale is the target's joint-HAC standard error (Appendix C); the daggered scale is OLS-derived. This bounds what may be claimed about the fit, and is not a specification test (§5.2). $d_{\mathcal{V}}$ is the number of forward-variance constraints—on SPX whatever VIX expiries were listed, on NDX the eight tenors of the corrected realised strip. The SPX two-stage protocol accepted its second stage at 2012, 2016, 2017 and 2018. The NDX forward-variance column should be read against the target's own 24% roughness (Section 5.5) rather than against zero: roughness co-moves strongly with residual size, which is consistent with target construction contributing materially but does not exclude kernel misspecification; the SPX column is subject to the ≈9.1% instrument bias of Section 5.3.

Multi-basin freedom, measured. The joint objective admits several basins and a poor seed can settle in one: seeding an off-index panel from a fit made against a different tenor set lands one date 41% worse on its own objective and the panel 5.0% worse. But the cold start is not one of the poor seeds, and the freedom among sensible ones is small. Holding the objective and the targets fixed and varying only the seed—cold, a warm continuation from the fit's own solution, and warm continuations from two independently converged fits—the total data cost spans 0.5094–0.5098, a range of 0.06%, with the *cold* start the best of the four; taking the per-date minimum across all four, which is in-sample selection and reported only as a bound, buys 0.8%—an order of magnitude smaller than the fit quality. What is loosely identified remains the parameter vector rather than the priced dynamics: any two fits the protocol accepts reproduce the same SSR and forward-variance term structures to within the reported RMS.

The identifiability spectrum, measured. Table 12 carries it. At each fitted $\boldsymbol{\theta}$ the *elasticity-scaled readout* Jacobian $J_{ij} = (\partial \mathcal{R}_i/\partial\theta_j)\,|\theta_j|/|\mathcal{R}_i|$ is formed over the seven free coordinates at the shipped configuration, with $\mathcal{R} = (\mathcal{S}, \mathcal{V})$, and its singular values recorded. It is the readout map that is differentiated, not the residual vector of (33): the objective weights and the ridge are excluded, so the spectrum describes what the *observables* determine rather than what the optimiser saw. Two readings matter. The conditioning is not uniform across regimes: the ratio $s_{\max}/s_{\min}$ runs from 353 to 5136, and the number of directions above 10^{-2} of the leading one—an effective rank at that tolerance—from two to six, median five out of seven. So the readout vector determines most but not all of the parameter vector, and how much depends on the year. The second reading is which coordinates sit in what is left over. Averaging the softest singular vector's loadings across the nine dates orders the parameters cleanly: ρ_{F} carries 0.712 of it and κ_{S} carries 0.018. The fast return correlation is the softest direction in the problem and the slow persistence is the stiffest—which is the reverse of what the parameters' *roles* would suggest, and is the reason a fitted ρ_{F} should not be read as a measurement.

Year	stage	$d_{\mathcal{R}}$	$s_{\max}$	$s_{\min}$	$s_{\max}/s_{\min}$	$\text{rank}_{10^{-2}}$
2012	warm	11	3.2	0.0080	403	6
2016	warm	14	88.8	0.0173	5136	2
2017	warm	15	83.5	0.0240	3474	3
2018	warm	14	17.6	0.0108	1635	5
2019	cold	14	3.0	0.0070	434	6
2020	cold	14	4.0	0.0096	410	5
2021	cold	17	8.5	0.0183	467	6
2022	cold	17	25.2	0.0713	353	5
2024	cold	17	50.6	0.0241	2095	4
median				0.0173	467	5

mean loading on the softest direction	ρ_{F}	ν_{S}	λ_{skew}	ν_{F}	ν_{L}	κ_{F}	κ_{S}
	0.712	0.426	0.218	0.183	0.107	0.092	0.018

Table 12: Elasticity-scaled readout-Jacobian spectrum at the nine shipped SPX fits, seven free coordinates—the readout map differentiated, objective weights and ridge excluded—generated from `jacobian_7free_ALL.json` by `emit_jacobian_table.py`. *stage* is which stage of the two-stage protocol that date's shipped fit came from. The lower block averages the softest singular vector's loadings over the nine dates, sorted.

Replication manifest. Table 13 lists the numerical constants the production path reads. It is not transcribed: it is emitted by `emit_manifest.py` directly from the objects the calibration constructs, or parsed from the fitter's own source, so it cannot drift from the engine. Values shown are the on-index defaults; off-index overrides are recorded per record. The same script writes `manifest.json`, carrying the box, per-date wall time, objective and Jacobian evaluation counts, the device, and for every record on both underlyings a truncated (12-hex-character) SHA-256 fingerprint and a reproduction check.

Each record states its own leverage-ladder configuration, so stage 2's source mollification (LAMH 4.0, PILLARAWARE 0) is legible from the artifact and not only from the protocol. That claim is checked rather than trusted: each record's own $\boldsymbol{\theta}$ is re-evaluated under both candidate ladders and compared elementwise against that record's own stored readout. On the device the fits were produced on, every one of the nine reproduces *bit-exactly* under the ladder it declares, and misses by 1.3–16.5% under the other—so the declaration and the numbers agree, by two independent routes. Off the device the fits were produced on, the same exercise reproduces each record's declared ladder but not its numbers exactly. On the one alternative device tested (CPU against MPS, both float32), the artifact records two quantities across the nine dates: the SSR-RMS gap runs 0.04–1.17 percentage points, and the mean relative elementwise deviation of the readout 0.14–0.62%. Those are observations on one device pair, not a bound for arbitrary hardware.

Code and data availability. The replication bundle is built by `make_release.py`, which walks the import graph from five entry points—the fitter `refit.py`, the two table generators, the leverage diagnostic and the smile-roll harness—and copies the resulting closure. It is 40 modules, not the six of the production path alone: those import path, band, smile and calibration helpers that a reader needs and the paper does not name. Two of the five need the option chains and so cannot be run from the bundle alone, `refit.py` and the smile-roll harness; the other three were checked by extracting the archive outside the source tree and running them there. Alongside the code the bundle carries the eighteen SPX and NDX fit records, the artifacts the tables are generated from (`manifest.json`, the readout-Jacobian spectrum, the leverage-remainder summaries), an environment lock, a `README`, and `SHA256SUMS` giving a full SHA-256 for every file shipped. It is not a copy of the working directory: superseded scripts, backup files, build output and the archived long-form appendix are absent by construction. Available from the authors on request. The option chains are licensed ORATS data and cannot be redistributed; Appendix D gives the filters and target constructions in enough detail to rebuild them from an equivalent source.

Setting	Value	Role
Quadrature and recompression		
n_{L}	5	within-step branch nodes
n_{X}	3	combined-factor quadrature
n_{P}	5	price sub-abscissas of the leverage
n_a	5	stationary factor grid, fast
n_b	5	stationary factor grid, slow
n_z	9	spot-shift grid for the smile readout
$n_{b,\mathsf{F}}$	5	recompression bands, fast
$n_{b,\mathsf{S}}$	3	recompression bands, slow
n_c	240	components carried per fibre (the budget)
n_q	7	variance-index quadrature
q_{vix}	5	leveraged variance-index quadrature
$z_{\max}$, Δt	0.12, 1/52	readout half-width, step
SSR tenors (weeks)	[1, 2, 4, 8, 13]	snapshot grid
Parameters		
free	7 of 8	ρ_{S} pinned at 0
solved, not fitted	$\bar{\gamma}$	from (18)
ν_{L} box	$[0.1, 3.0]$	widened from the default
log-variance clamp	$[-40, 12]$	before exponentiation; bounds V_ℓ
floors ℓ, V	10^{-6}, 10^{-16}	leverage and variance
Objective and optimiser		
w_{vov}	0.8	forward-variance block weight
ς	1 (SPX), inverse rel. HAC s.e. (NDX)	per-tenor SSR weights
ridge ϱ	0.03 (ν, λ), 0.15 (ρ, κ)	relative, toward the stage anchor
ridge floor ε	0.1	in $\max(\lvert\theta_k^0\rvert, \varepsilon)$
ftol, xtol	10^{-6}, 10^{-6}	least-squares stopping
max nfev	40	per stage
Protocol		
stage 1	cold	LADDER 42, VOVLEV 1, λ avg, PIN ρ_{S}=0
stage 2	warm from stage 1	LAMH 4.0, PILLARAWARE 0
acceptance	per date	stage 2 only where its data cost is lower
calendar interpolation	linear in T (BLEND)	between listed expiries
VIXFIX, VOVLAMTEN	1, avg	expiry expansion; λ window rule
Environment and determinism		
device, dtype	MPS, float32	readouts are device-sensitive
torch, python	2.10.0, 3.14.0	
random seeds	none	quadrature and optimiser are deterministic

Table 13: The *principal* production configuration—the constants a reader must set to reproduce the fits, not an exhaustive list of every literal in the path (the skew-difference step 6×10^{-3}, the eight Newton iterations of the at-the-money inversion and the 60-step VIX bisection are fixed in the code and not exposed)—emitted by `emit_manifest.py`. The quadrature counts, the box, the clamps and the optimiser settings are read from the objects the calibration constructs or parsed from the fitter's own source, so none of them is transcribed.

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